13

Normal Labor and Delivery

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OUTLINE

KEY ABBREVIATIONS

American Academy of Pediatrics	AAP
American College of Obstetricians and Gynecologists	ACOG
Body mass index	BMI
Cephalopelvic disproportion	CPD
Cervical length	CL
Cesarean delivery	CD
Computed tomography	CT
Dehydroepiandrosterone sulfate	DHEAS
Fetal heart rate	FHR
Intrauterine pressure catheter	IUPC
Intraventricular hemorrhage	IVH
Left occiput anterior	LOA
Magnetic resonance imaging	MRI
Mean difference	MD
Montevideo unit	MVU
Normal saline	NS
Occiput anterior	OA
Occiput posterior	OP
Occiput transverse	OT
Prostaglandin	PG
Randomized controlled trial	RCT
Right occiput anterior	ROA
Skin-to-skin contact	SSC

OVERVIEW

The initiation of normal labor at term requires endocrine, paracrine, and autocrine signaling among the fetus, uterus, placenta, and mother. Although the exact trigger for human labor at term remains unknown, it is believed to involve conversion of fetal dehydroepiandrosterone sulfate (DHEAS) to estriol and estradiol by the placenta. These hormones upregulate transcription of progesterone, progesterone receptors, oxytocin receptors, and gap junction proteins within the uterus, which helps to facilitate regular uterine contractions. The *latent phase* of labor is characterized by a slower rate of cervical dilation, whereas the *active phase* of labor is characterized by a faster rate of cervical dilation and does not begin for most women until the cervix is dilated 6 cm. The duration of the second stage of labor can be affected by a number of factors, including epidural use, fetal position, fetal weight, ethnicity, and parity. This chapter reviews the characteristics and physiology of normal labor at term. Factors that affect the average duration of progress of the first and second stages of labor will be reviewed, and an evidence-based evaluation of strategies to support the mother during labor and facilitate safe delivery of the fetus will be presented.

LABOR: DEFINITION AND PHYSIOLOGY

***Labor* is defined as the process by which the fetus is expelled from the uterus. More specifically, labor requires regular, effective**

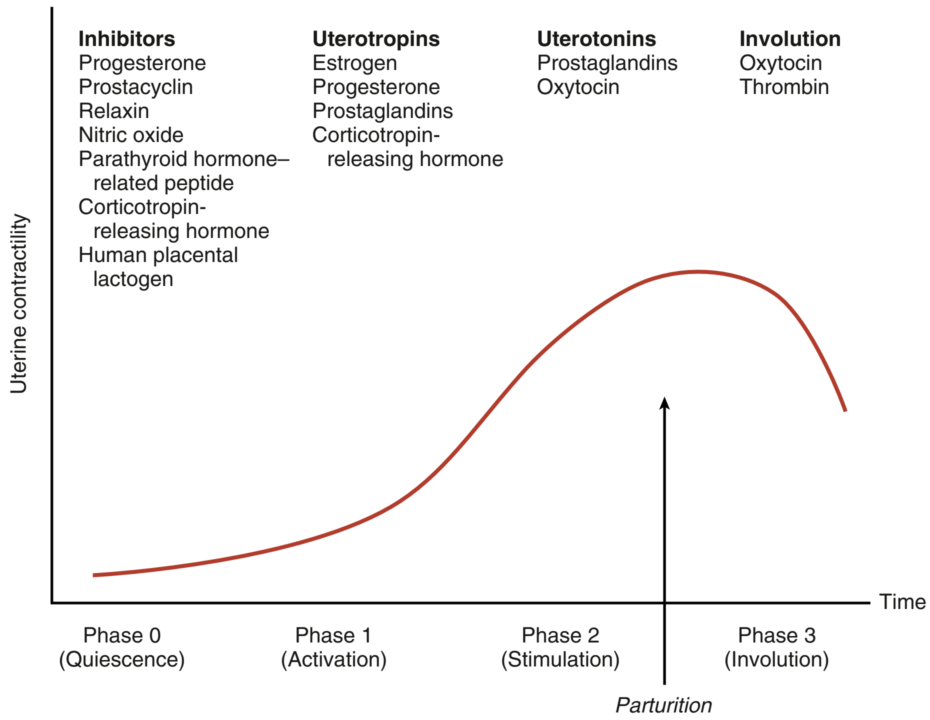


Fig. 13.1 **Regulation of Uterine Activity During Pregnancy and Labor.** (Modified from Challis JRG, Gibb W. Control of parturition. *Prenat Neonat Med.* 1996;1:283.)

contractions that lead to dilation and effacement of the cervix. This chapter describes the physiology and normal characteristics of term labor and delivery.

The physiology of labor initiation has not been completely elucidated, but the putative mechanisms have been well reviewed by Liao and colleagues.[1] Labor initiation is species specific, and the mechanisms of human labor are unique. **The four phases of labor from quiescence to involution are outlined in** Fig. 13.1.[2] The first phase is **quiescence,** which represents that time in utero before labor begins, when uterine activity is suppressed by the action of progesterone, prostacyclin, relaxin, nitric oxide, parathyroid hormone–related peptide, and possibly other hormones. During the **activation phase,** estrogen begins to facilitate expression of myometrial receptors for prostaglandins (PGs) and oxytocin, which results in ion channel activation and increased gap junctions. This increase in the gap junctions between myometrial cells facilitates effective contractions.[3] In essence, the activation phase readies the uterus for the subsequent **stimulation phase,** when uterotonics—particularly PGs and oxytocin—stimulate regular contractions. In humans, this process at term may be protracted, occurring over days to weeks. The final phase, **uterine involution,** occurs after delivery and is mediated primarily by oxytocin. The first three phases of labor require endocrine, paracrine, and autocrine interaction among the fetus, membranes, placenta, and mother.

The fetus has a central role in the initiation of term labor in nonhuman mammals; in humans, the fetal role is not completely understood (Fig. 13.2).[2-5] In sheep, term labor is initiated through activation of the fetal hypothalamic-pituitary-adrenal axis, with a resultant increase in fetal adrenocorticotropic hormone and cortisol.[4,5] Fetal cortisol increases production of estradiol and decreases production of progesterone by a shift in placental metabolism of cortisol dependent on placental 17α-hydroxylase. The change in the circulating progesterone/estradiol concentration stimulates placental production of oxytocin and PGs, particularly prostaglandin $F_{2\alpha}$ ($PGF_{2\alpha}$), which in turn promotes myometrial contractility.[4] If this increase in fetal adrenocorticotropic hormone and cortisol is blocked, progesterone levels remain unchanged, and parturition is delayed.[5] In contrast, humans lack placental 17α-hydroxylase, maternal and fetal levels of progesterone remain elevated, and no trigger exists for parturition because of an increase in fetal cortisol near term. Rather, in humans, evidence suggests that placental production of corticotropin-releasing hormone near term activates the fetal hypothalamic-pituitary axis and results in increased production of dehydroepiandrostenedione by the fetal adrenal gland.[6] Fetal dehydroepiandrostenedione is converted in the placenta to estradiol and estriol. Placenta-derived estriol potentiates uterine activity by enhancing the transcription of maternal (likely decidual) $PGF_{2\alpha}$, PG receptors, oxytocin receptors, and gap junction proteins.[6-9] In humans, no documented decrease in progesterone has been observed near term, and a fall in progesterone is not necessary for labor initiation. However, some research suggests the possibility of a functional progesterone withdrawal in humans. Labor is accompanied by a decrease in the concentration of progesterone receptors and a change in the ratio of progesterone receptor isoforms A and B in both the myometrium[10-13] and the membranes.[13] During labor, increased expression of nuclear and membrane progesterone receptor isoforms serves to enhance genomic expression of contraction-associated proteins, increase intracellular calcium, and decrease cyclic adenosine monophosphate (cAMP).[14] More research is needed to elucidate the precise mechanism through which the human parturition cascade is activated. Fetal maturation might play an important role, as might maternal cues that affect circadian cycling. Most species have distinct diurnal patterns of contractions and delivery, and in humans, the majority of contractions occur at night.[2,15]

Oxytocin is commonly used for labor induction and augmentation, and a full understanding of the mechanism of oxytocin action is

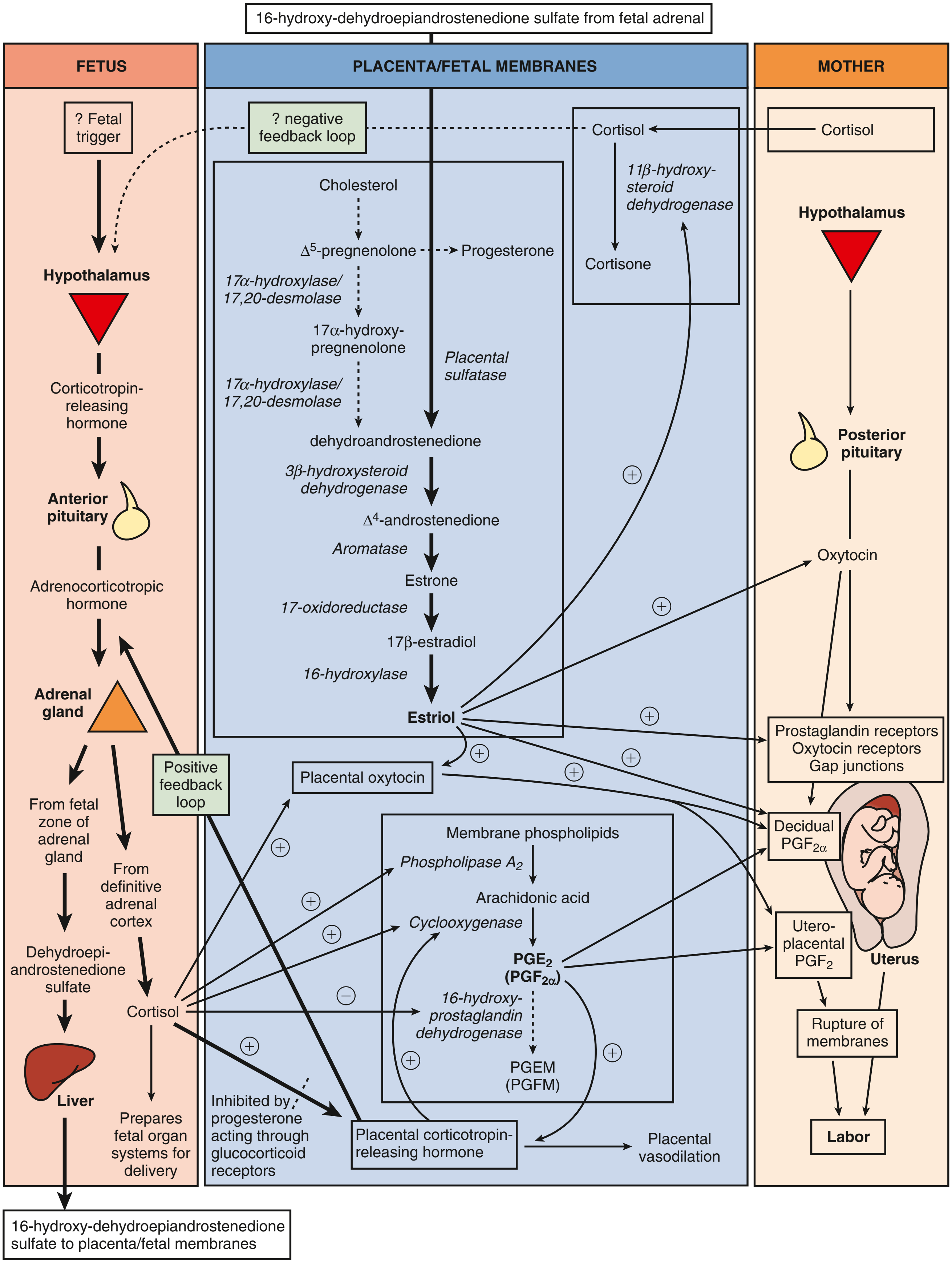


Fig. 13.2 Proposed "Parturition Cascade" for Labor Induction at Term. Spontaneous induction of labor at term in the human is regulated by a series of paracrine/autocrine hormones acting in an integrated parturition cascade responsible for promoting uterine contractions. *PGE₂*, Prostaglandin E_2; *PGEM*, 13,14-dihydro-15-keto-PGE_2; *PGF_2*, prostaglandin F_2; *$PGF_{2\alpha}$*, prostaglandin $F_{2\alpha}$; *PGFM*, 13,14-dihydro-15-keto-$PGF_{2\alpha}$. (Modified from Norwitz ER, Robinson JN, Repke JT, et al. The initiation of parturition: a comparative analysis across the species. *Curr Prob Obstet Gynecol Fertil.* 1999;22:44-71.)

important. **Oxytocin is a peptide hormone synthesized in the hypothalamus and released from the posterior pituitary in a pulsatile fashion. At term, oxytocin serves as a potent uterotonic agent capable of stimulating uterine contractions at intravenous (IV) infusion rates of 1 to 2 mIU/min.[16] Oxytocin is inactivated largely in the liver and kidney; during pregnancy, it is degraded primarily by placental oxytocinase. Its biological half-life is approximately 3 to 4 minutes, but it appears to be shorter when higher doses are infused.** Concentrations of oxytocin in the maternal circulation do not change significantly during pregnancy or before the onset of labor, but they do rise late in the second stage of labor.[16,17] Studies of fetal pituitary oxytocin production and the umbilical arteriovenous differences in plasma oxytocin strongly suggest that the fetus secretes oxytocin that reaches the maternal side of the placenta.[16,18] **The calculated rate of active oxytocin secretion from the fetus increases from a baseline of 1 mIU/min before labor to around 3 mIU/min after spontaneous labor.[18]**

Significant differences in myometrial oxytocin receptor distribution have been reported, with large numbers of fundal receptors and fewer receptors in the lower uterine segment and cervix.[19] Myometrial oxytocin receptors increase on average by 100- to 200-fold during pregnancy and reach a maximum during early labor.[16,17,20,21] This rise in receptor concentration is paralleled by an increase in uterine sensitivity to circulating oxytocin. Specific high-affinity oxytocin receptors have also been isolated from human amnion and decidua parietalis but not decidua vera.[16,19] **It has been suggested that oxytocin plays a dual role in parturition. First, through its receptor, oxytocin directly stimulates uterine contractions. Second, oxytocin may act indirectly by stimulating the amnion and decidua to produce PGs.[19,22-24] Indeed, even when uterine contractions are adequate, induction of labor at term is successful only when oxytocin infusion is associated with an increase in PGF production.[19]**

Oxytocin binding to its receptor activates phospholipase C.[25] In turn, phospholipase C increases intracellular calcium both by stimulating the release of intracellular calcium and by promoting the influx of extracellular calcium. Oxytocin stimulation of phospholipase C can be inhibited by increased levels of cAMP.[25] Increased calcium levels stimulate the calmodulin-mediated activation of myosin light-chain kinase. Oxytocin may also stimulate uterine contractions via a calcium-independent pathway by inhibiting myosin phosphatase, which in turn increases myosin phosphorylation. **These pathways (of phospholipase A_2 and intracellular calcium) have been the target of multiple tocolytic agents: indomethacin, calcium channel blockers, β-mimetics (through stimulation of cAMP), and magnesium.**

MECHANICS OF LABOR

Labor and delivery are active processes. The ability of the fetus to successfully negotiate the pelvis during labor and delivery depends on the complex interactions of three variables: uterine activity, the fetus, and the maternal pelvis. This complex relationship has been simplified in the mnemonic *powers, passenger, passage*.

Uterine Activity (Powers)

The *powers* refer to the forces generated by the uterine musculature. Uterine activity is characterized by the frequency, amplitude (intensity), and duration of contractions. Assessment of uterine activity may include simple observation, manual palpation, external objective assessment techniques (such as external tocodynamometry), and direct measurement via an intrauterine pressure catheter (IUPC). External tocodynamometry measures the change in shape of the abdominal wall as a function of uterine contractions and, as such, is qualitative rather than quantitative. Although it permits graphic display of uterine activity and allows for accurate correlation of fetal heart rate (FHR) patterns with uterine activity, external tocodynamometry does not allow measurement of contraction intensity or basal intrauterine tone. The most precise method for determination of uterine activity is the direct measurement of intrauterine pressure with an IUPC. However, this procedure should not be performed unless indicated given the small but finite associated risk of maternal fever.[26] Several historical case reports documented additional complications, including placental disruption, uterine perforation, and fetal-maternal hemorrhage, which were associated with first-generation versions of this device.[27,28]

Despite technologic improvements, the definition of "adequate" uterine activity during labor remains unclear. Classically, three to five contractions in 10 minutes has been used to define adequate labor; this pattern has been observed in approximately 95% of women in spontaneous labor. In labor, patients usually contract every 2 to 5 minutes, with contractions becoming as frequent as every 2 to 3 minutes in late active labor and during the second stage. Abnormal uterine activity can also be observed either spontaneously or as a result of iatrogenic interventions. *Tachysystole* is defined as more than five contractions in 10 minutes averaged over 30 minutes. If tachysystole occurs, documentation should note the presence or absence of FHR decelerations. The term *hyperstimulation* should no longer be used.[29]

Various units of measure have been devised to objectively quantify uterine activity, the most common of which is the ***Montevideo unit* (MVU),** a measure of average frequency and amplitude above basal tone (the average strength of contractions in millimeters of mercury multiplied by the number of contractions per 10 minutes). Although 150 to 350 MVUs has been described for adequate labor, 200 to 250 MVUs is commonly accepted to define adequate labor in the active phase.[29,30] No data identify adequate forces during latent labor. Although it is generally believed that optimal uterine contractions are associated with an increased likelihood of vaginal delivery, data to support this assumption are limited. **If uterine contractions are "adequate" to effect vaginal delivery, one of two things will happen: either the cervix will efface and dilate and the fetal head will descend, or caput succedaneum (scalp edema) and molding of the fetal head (overlapping of the skull bones) will worsen without cervical effacement and dilation. The latter situation suggests the presence of cephalopelvic disproportion (CPD),** which can be either *absolute,* in which the fetus is simply too large to negotiate the pelvis, or *relative,* in which delivery of the fetus through the pelvis would be possible under optimal conditions but is precluded by malposition or abnormal attitude of the fetal head.

Fetus (Passenger)

The passenger, of course, is the fetus. Several fetal variables influence the course of labor and delivery. **Fetal size** can be estimated clinically by abdominal palpation or ultrasound or by asking a multiparous patient about her best estimate, but all of these methods are subject to a large degree of error. ***Fetal macrosomia* is defined by the American College of Obstetricians and Gynecologists (ACOG) as birthweight greater than 4000 to 4500 g for any gestational age,[31]** and it is associated with an increased likelihood of cesarean delivery (CD), labor dystocia, CD after a failed trial of labor, shoulder dystocia, and birth trauma.[32] **Fetal *lie* refers to the longitudinal axis of the fetus relative to the longitudinal axis of the uterus.** Fetal lie can be longitudinal, transverse, or oblique (Fig. 13.3). In a singleton pregnancy, only fetuses in a longitudinal lie can be safely delivered vaginally.

Presentation* refers to the fetal part that directly overlies the pelvic inlet.** In a fetus presenting in the longitudinal lie, the presentation can be cephalic or breech. *Compound presentation* refers to the presence of more than one fetal part overlying the pelvic inlet, such as a fetal hand and the vertex. *Funic presentation* refers to presentation of the umbilical cord and is rare at term. In a cephalic fetus, the presentation is classified according to the leading bony landmark of the skull, which can be either the occiput (vertex), the chin (mentum), or the brow (Fig. 13.4). ***Malpresentation, a term that refers to any presentation other than vertex, is seen in approximately 5% of all term labors (see Chapter 19).

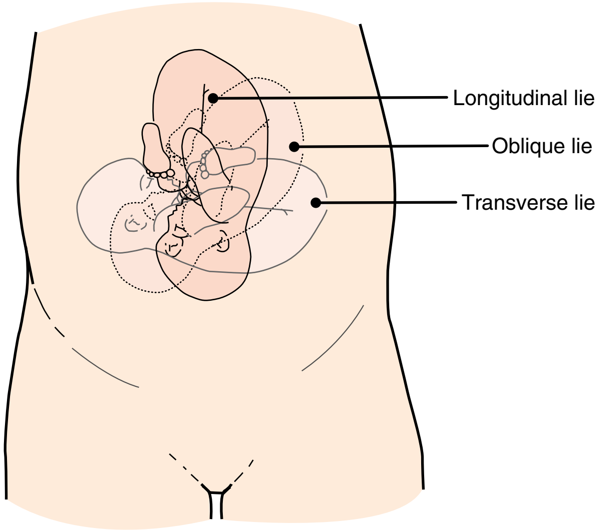


Fig. 13.3 Examples of Fetal Lie.

Attitude refers to the position of the head with regard to the fetal spine (the degree of flexion and/or extension of the fetal head). Flexion of the head is important to facilitate *engagement* of the head in the maternal pelvis. When the fetal chin is optimally flexed onto the chest, the suboccipitobregmatic diameter (9.5 cm) presents at the pelvic inlet (Fig. 13.5). This is the smallest possible presenting diameter in the cephalic presentation. As the head deflexes (extends), the diameter presenting to the pelvic inlet progressively increases even before the malpresentations of brow and face are encountered (see Fig. 13.5) and may contribute to failure to progress in labor. The architecture of the pelvic floor along with increased uterine activity may correct deflexion in the early stages of labor.

Position of the fetus refers to the relationship of the fetal presenting part to the maternal pelvis, and it can be assessed by vaginal or ultrasound examination. (For discussion of the data utilizing ultrasound for determination of fetal position in labor, see Ultrasound in Labor and Delivery section below). For *cephalic presentations,* the fetal occiput is the reference: if the occiput is directly anterior, the position is occiput anterior (OA); if the occiput is turned toward the mother's right side, the position is right occiput anterior (ROA). In the *breech presentation,* the sacrum is the reference (right sacrum anterior). The various positions of a cephalic presentation are illustrated in Fig. 13.6.

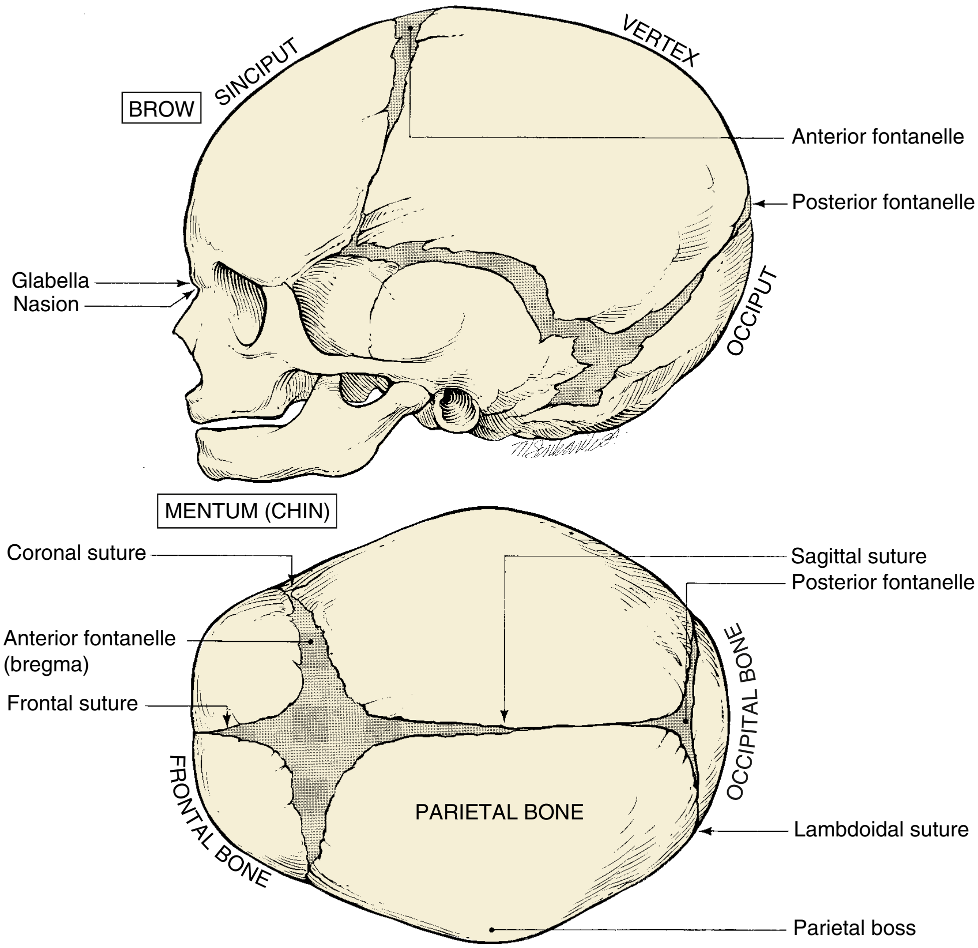


Fig. 13.4 Landmarks of Fetal Skull for Determination of Fetal Position.

In a *vertex presentation,* position can be determined by palpation of the fetal sutures. The sagittal suture is the easiest to palpate, but palpation of the distinctive lambdoid sutures should identify the position of the fetal occiput; the frontal suture can also be used to determine the position of the front of the vertex.

Most commonly, the fetal head enters the pelvis in a transverse position and then, as a normal part of labor, it rotates to an OA position. Most fetuses deliver in the OA, left occiput anterior (LOA), or ROA position. ***Malposition*** refers to any position in labor that is not in the above three categories. In the past, fewer than 10% of presentations were occiput posterior (OP) at delivery.[33] However, epidural analgesia may be an independent risk factor for persistent OP presentation in labor. In an observational cohort study, OP presentation was observed in 12.9% of women with epidurals compared with 3.3% of controls (P = .002).[34] In a Cochrane meta-analysis of four randomized controlled trials (RCTs), malposition was 40% more likely for women with an epidural compared with controls; however, this difference was not statistically significant, and more RCTs are needed (odds ratio [OR], 1.40; 95% confidence interval [CI], 0.98 to 1.99).[35] ***Asynclitism*** occurs when the sagittal suture is not directly central relative to the maternal pelvis. If the fetal head is turned such that more parietal bone is present posteriorly, the sagittal suture is more anterior; this is referred to as *posterior asynclitism.* In contrast, *anterior asynclitism* occurs as more parietal bone presents anteriorly.[36] The occiput transverse (OT) and OP positions are less common at delivery and are more difficult to deliver vaginally.

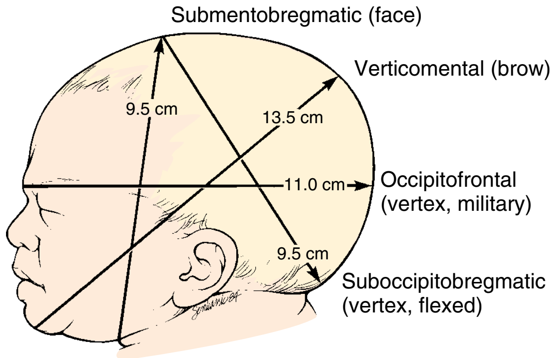


Fig. 13.5 Presenting Diameters of the Average Term Fetal Skull.

Station **is a measure of descent of the bony presenting part of the fetus through the birth canal** (Fig. 13.7). The current standard classification (−5 to +5) is based on a quantitative measure in centimeters of the distance of the leading bony edge from the ischial spines. The *midpoint* (0 station) is defined as the plane of the maternal ischial spines. The ischial spines can be palpated on vaginal examination at approximately 8 o'clock and 4 o'clock. For the right-handed person, they are most easily felt on the maternal right.

An abnormality in any of these fetal variables may affect both the course of labor and the route of delivery. For example, OP presentation is well known to be associated with longer labor, operative vaginal delivery (OVD), and an increased risk of CD.[34,37]

Maternal Pelvis (Passage)

The passage consists of the bony pelvis—composed of the sacrum, ilium, ischium, and pubis—and the resistance provided by the soft

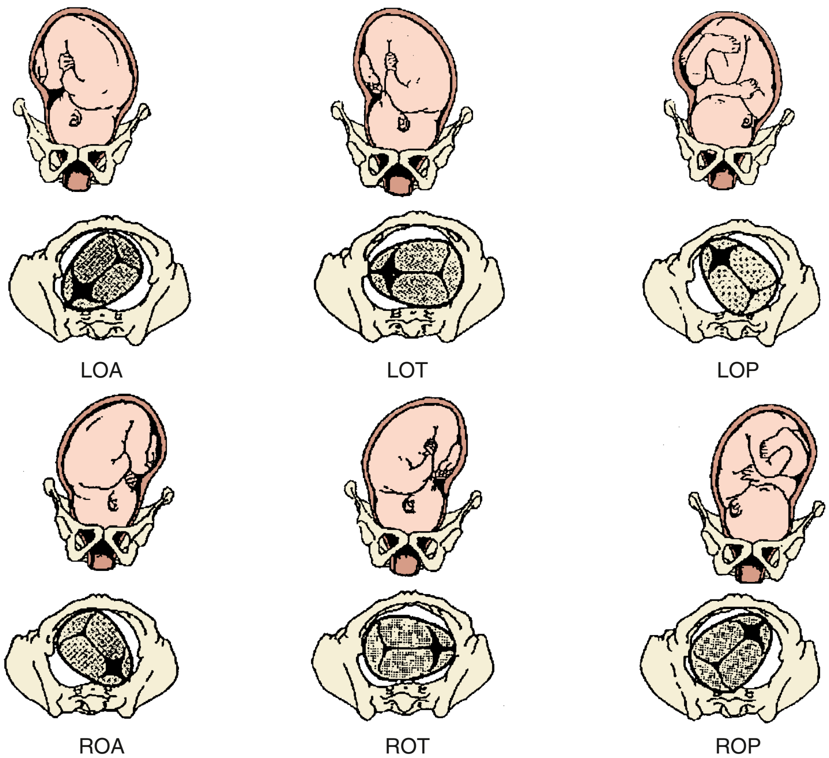


Fig. 13.6 **Fetal Presentations and Positions in Labor.** *LOA,* Left occiput anterior; *LOP,* left occiput posterior; *LOT,* left occiput transverse; *ROA,* right occiput anterior; *ROP,* right occiput posterior; *ROT,* right occiput transverse. (Modified from Norwitz ER, Robinson J, Repke JT, et al. The initiation and management of labor. In: Seifer DB, Samuels P, Kniss DA, eds. *The Physiologic Basis of Gynecology and Obstetrics.* Philadelphia: Lippincott Williams & Wilkins; 2001.)

tissues. The bony pelvis is divided into the *false* (greater) and *true* (lesser) pelvis by the pelvic brim, which is demarcated by the sacral promontory, the anterior ala of the sacrum, the arcuate line of the ilium, the pectineal line of the pubis, and the pubic crest culminating in the symphysis (Fig. 13.8). Measurements of the various parameters of the bony female pelvis have been made with great precision, directly in cadavers and using radiographic imaging in living women. Such measurements have divided the true pelvis into a series of planes that must be negotiated by the fetus during passage through the birth canal, which can be broadly termed the *pelvic inlet, midpelvis,* and *pelvic outlet.* Pelvimetry performed with radiographic computed tomography (CT) or magnetic resonance imaging (MRI) has been used to determine average and critical limit values for the various parameters of the bony pelvis.[38,39] Critical limit values are measurements that may be associated with a significant probability of CPD depending upon fetal size and gestational age. However, subsequent studies were unable to demonstrate threshold pelvic or fetal cutoff values with sufficient sensitivity or specificity to predict CPD and the subsequent need for CD prior to the onset of labor.[40,41] In current obstetric practice, radiographic CT and MRI pelvimetry are rarely used given the lack of evidence of benefit and some data that show possible harm (increased incidence of CD); instead, a clinical trial of the pelvis (labor) is used. The remaining indications for radiography, CT pelvimetry, or MRI are evaluation for vaginal breech delivery or evaluation of a woman who has suffered a significant pelvic fracture.[42]

Clinical pelvimetry is currently the only method of assessing the shape and dimensions of the bony pelvis in labor[40] (Video 13.1). A useful protocol for clinical pelvimetry is detailed in Fig. 13.9 and involves assessment of the pelvic inlet, midpelvis, and pelvic outlet. Reported average and critical limit pelvic diameters may be used as a historical reference during the clinical examination to determine pelvic shape and assess risk for CPD. The inlet of the true pelvis is largest in its transverse diameter and averages 13.5 cm.[40] The *diagonal conjugate,* the distance from the sacral promontory to the inferior margin of the symphysis pubis as assessed on vaginal examination, is a clinical representation of the anteroposterior (AP) diameter of the pelvic inlet. The *true conjugate,* or *obstetric conjugate,* of the pelvic inlet is the distance from the sacral promontory to the superior aspect of the symphysis pubis. The obstetric conjugate has an average value of

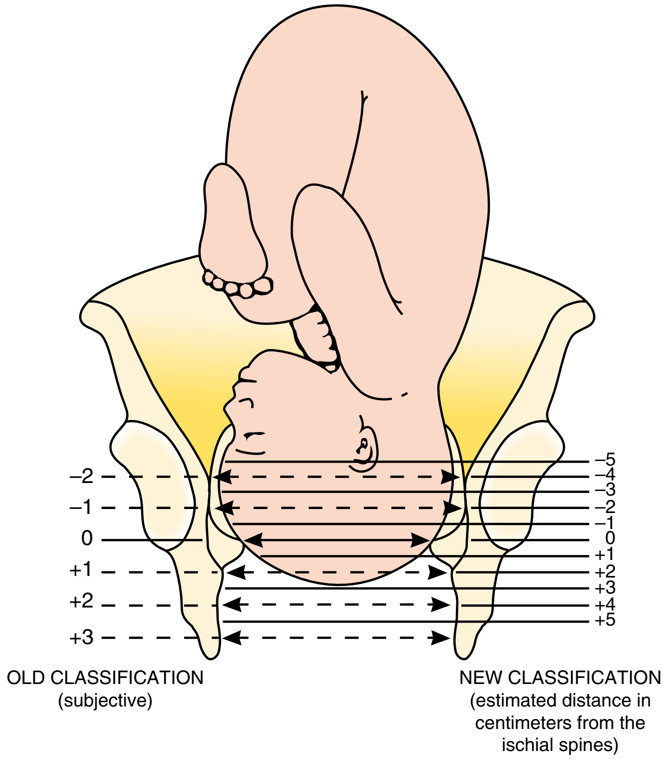


Fig. 13.7 Relationship of the Leading Edge of the Presenting Part of the Fetus to the Plane of the Maternal Ischial Spines Determines the Station. Station +1/+3 (old classification), or +2/+5 (new classification), is illustrated.

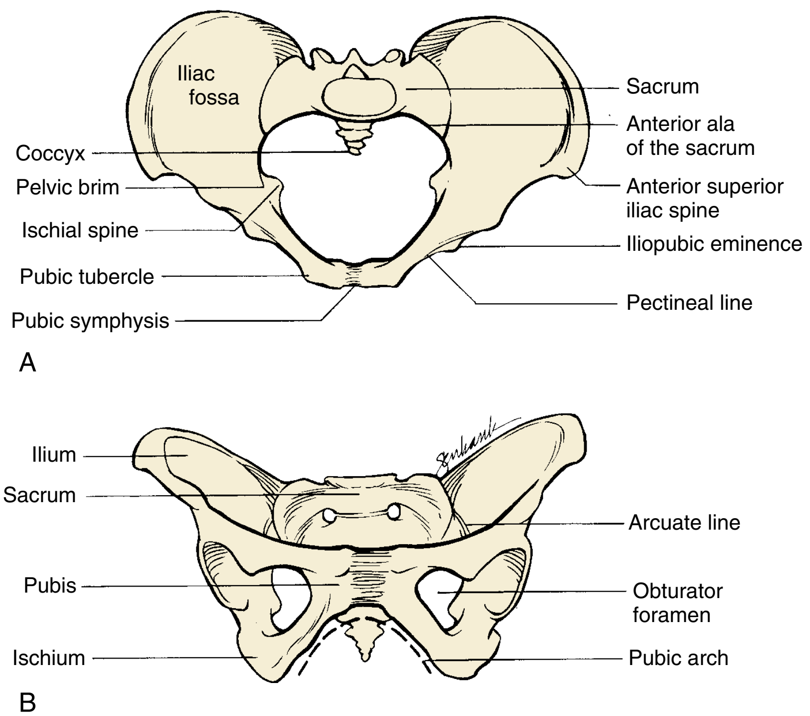


Fig. 13.8 Superior (A) and Anterior (B) Views of the Female Pelvis. (From Repke JT. *Intrapartum Obstetrics.* New York: Churchill Livingstone; 1996:68.)

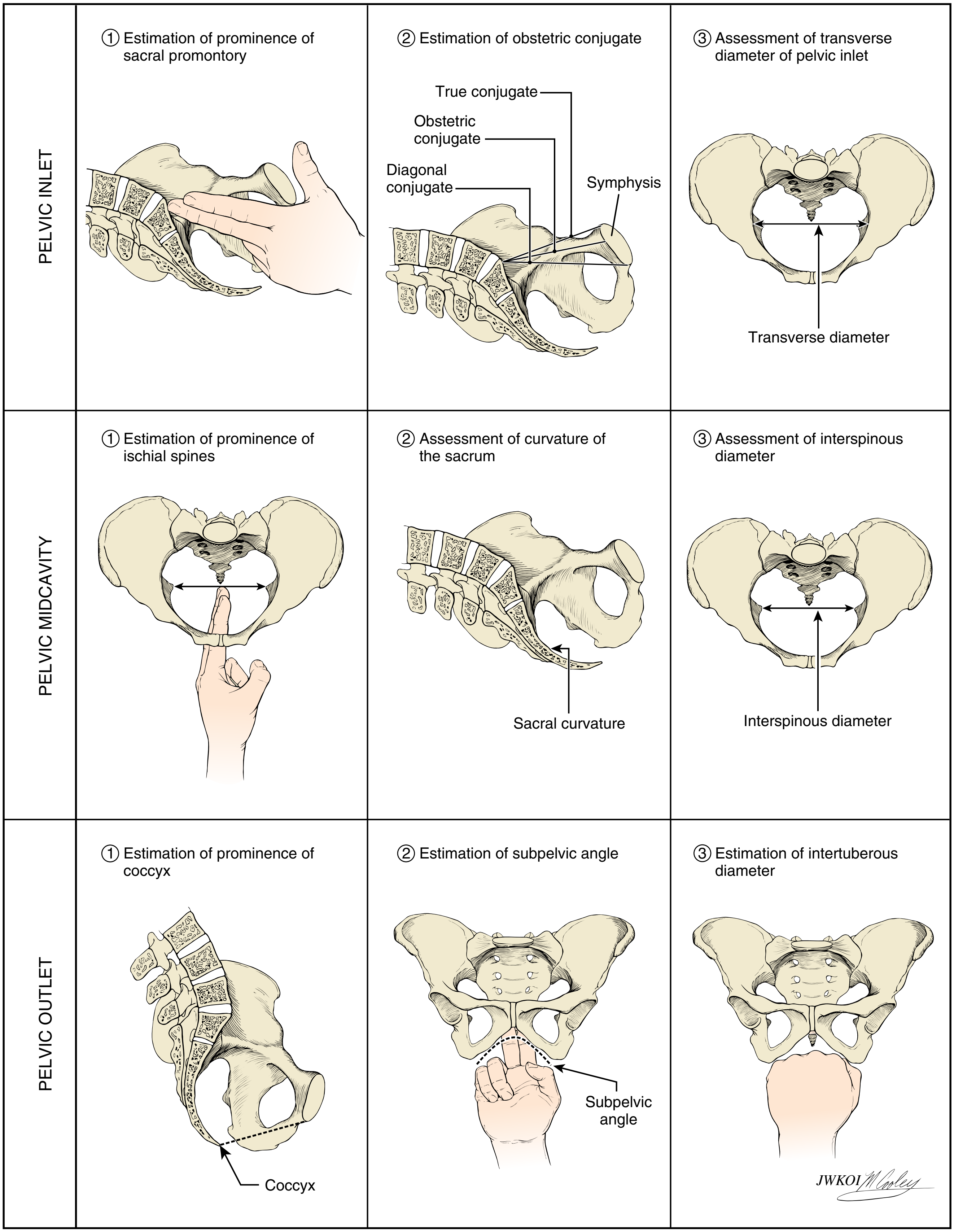


Fig. 13.9 Protocol for Clinical Pelvimetry.

11 cm and is the smallest diameter of the inlet. It is considered to be contracted if it measures less than 10 cm.[40] The obstetric conjugate cannot be measured clinically but can be estimated by subtracting 1.5 to 2.0 cm from the diagonal conjugate, which has an average distance of 12.5 cm.

The limiting factor in the midpelvis is the transverse interspinous diameter (the measurement between the ischial spines), which is usually the smallest diameter of the pelvis but should be greater than 10 cm. The pelvic outlet is rarely of clinical significance, however. The average pubic angle is greater than 90 degrees and will typically accommodate two fingerbreadths.[40] The AP diameter from the coccyx to the symphysis pubis is approximately 13 cm in most cases, and the transverse diameter between the ischial tuberosities is approximately 8 cm and will typically accommodate four knuckles (see Fig. 13.9).

The shape of the female bony pelvis can be classified into four broad categories: gynecoid, anthropoid, android, and platypelloid (Fig. 13.10). This classification is based on the radiographic studies of Caldwell and Moloy[43] and separates those pelvic shapes with more favorable characteristics (gynecoid, anthropoid) from those less favorable for vaginal delivery (android, platypelloid). In reality, however, many women fall into intermediate classes, and the distinctions become arbitrary. The *gynecoid* pelvis is the classic female shape. The *anthropoid* pelvis—with its exaggerated oval shape of the inlet, largest AP diameter, and limited anterior capacity—is more often associated with delivery in the OP position. The *android* pelvis is male in pattern and theoretically has an increased risk of CPD, and the broad and flat *platypelloid* pelvis theoretically predisposes to a transverse arrest. Although the assessment of fetal size, along with pelvic shape and capacity, is still of clinical utility, it is a very inexact science. An adequate trial of labor is the only definitive method to determine whether a fetus will be able to safely negotiate through the pelvis.

Pelvic soft tissues may provide resistance in both the first and second stages of labor. In the first stage, resistance is offered primarily by the cervix, whereas in the second stage it is offered by the muscles of the pelvic floor. In the second stage of labor, the resistance of the pelvic musculature is believed to play an important role in the rotation and movement of the presenting part through the pelvis.

CARDINAL MOVEMENTS IN LABOR

The *cardinal movements* refer to changes in the position of the fetal head during its passage through the birth canal. Because of the asymmetry of the shape of both the fetal head and the maternal bony pelvis, such rotations are required for the fetus to successfully negotiate the birth canal. **Although labor and birth comprise a continuous process, seven discrete cardinal movements are described: (1) engagement, (2) descent, (3) flexion, (4) internal rotation, (5) extension, (6) external rotation or restitution, and (7) expulsion** (Fig. 13.11; Video 13.2).

Engagement

***Engagement* refers to passage of the widest diameter of the presenting part to a level below the plane of the pelvic inlet** (Fig. 13.12). In cephalic presentation with a well-flexed head, the largest transverse

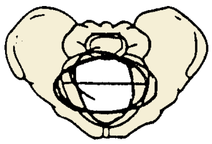
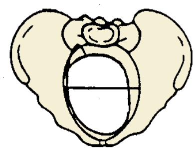
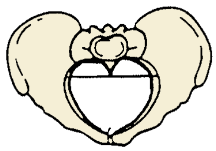
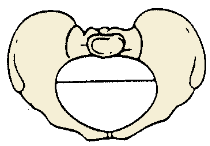
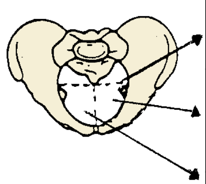
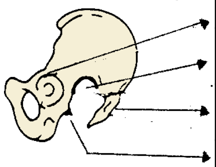
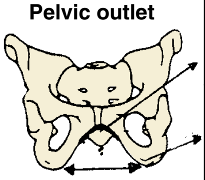
		 Gynecoid	 Anthropoid	 Android	 Platypelloid
Pelvic inlet 	Widest transverse diameter of inlet	12 cm	<12 cm	12 cm	12 cm
	Anteroposterior diameter of inlet	11 cm	>12 cm	11 cm	10 cm
	Forepelvis	Wide	Divergent	Narrow	Straight
Pelvic midcavity 	Side walls	Straight	Narrow	Convergent	Wide
	Sacrosciatic notch	Medium	Backward	Narrow	Forward
	Inclination of sacrum	Medium	Wide	Forward (lower third)	Narrow
	Ischial spines	Not prominent	Not prominent	Not prominent	Not prominent
Pelvic outlet 	Subpubic arch	Wide	Medium	Narrow	Wide
	Transverse diameter of outlet	10 cm	10 cm	<10 cm	10 cm

Fig. 13.10 Characteristics of the Four Types of Female Bony Pelvis. (Modified from Callahan TL, Caughey AB, Heffner LJ, eds. *Blueprints in Obstetrics and Gynecology*. Malden, MA: Blackwell Science; 1998:45.)

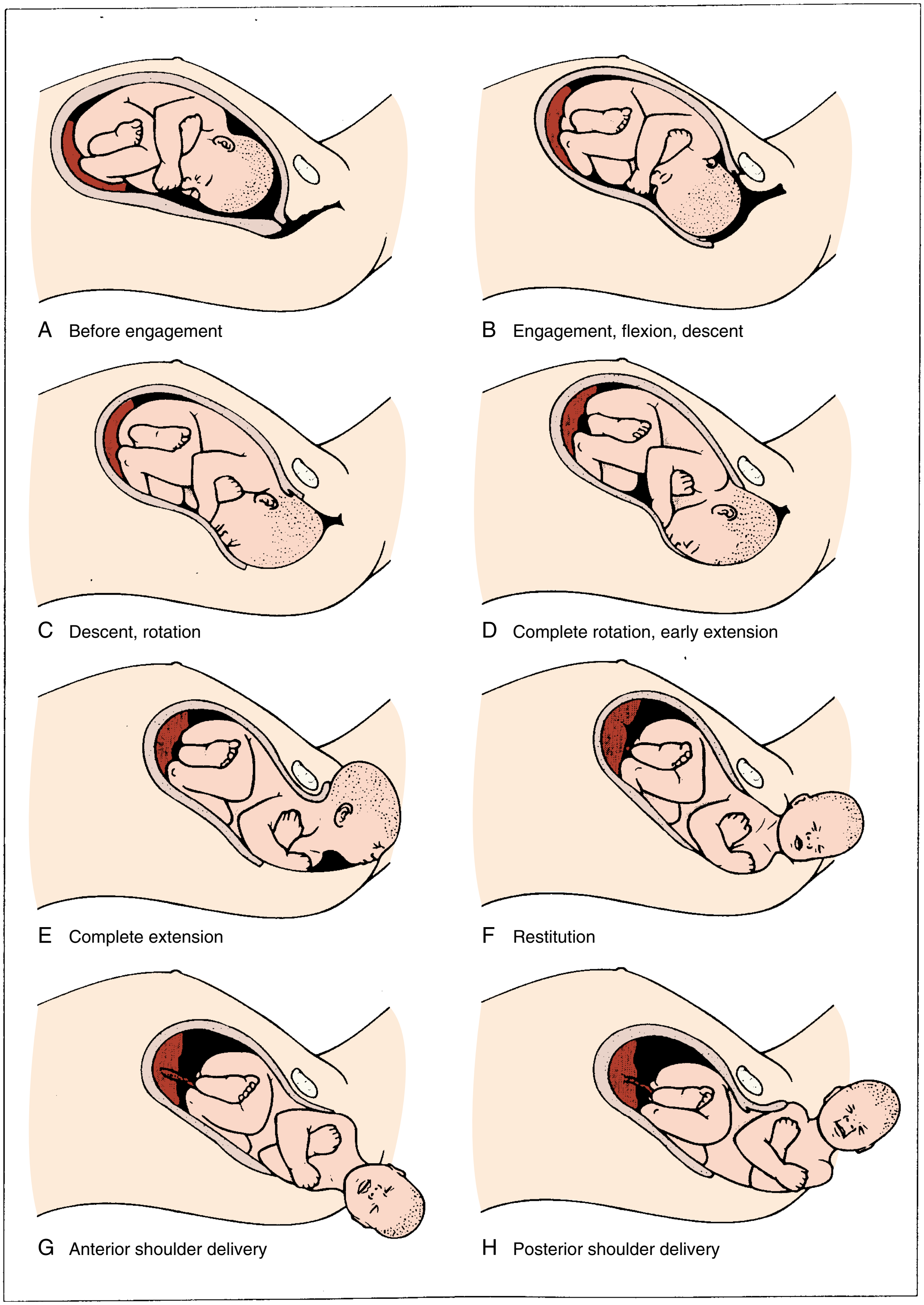


Fig. 13.11 Cardinal Movements of Labor.

diameter of the fetal head is the biparietal diameter (9.5 cm). In breech presentation the widest diameter is the bitrochanteric diameter. Clinically, engagement can be confirmed by palpation of the presenting part both abdominally and vaginally. **With a cephalic presentation, engagement is achieved when the leading bony presenting part is at 0 station on vaginal examination.**[44,45] Engagement is considered an important clinical prognostic sign because it demonstrates that, at least at the level of the pelvic inlet, the maternal bony

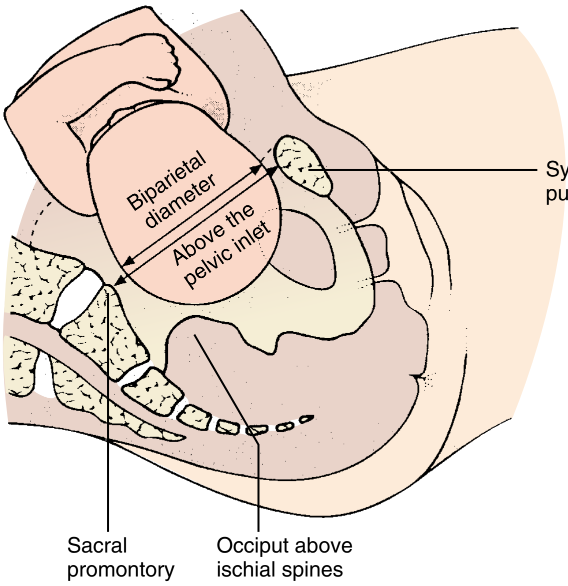


Fig. 13.12 Engagement of the Fetal Head.

pelvis is sufficiently large to allow descent of the fetal head. In nulliparas, engagement of the fetal head usually occurs by 36 weeks' gestation; however, in multiparas engagement can occur later in gestation or even during the course of labor.

Descent

Descent refers to the downward passage of the presenting part through the pelvis. Descent of the fetus is not continuous; the greatest rates of descent occur in the late active phase and during the second stage of labor.

Flexion

Flexion of the fetal head occurs passively as the head descends, owing to the shape of the bony pelvis and the resistance offered by the soft tissues of the pelvic floor. Although flexion of the fetal head onto the chest is present to some degree in most fetuses before labor, complete flexion usually occurs only during the course of labor. The result of complete flexion is to present the smallest diameter of the fetal head (the suboccipitobregmatic diameter) for optimal passage through the pelvis.

Internal Rotation

Internal rotation refers to rotation of the presenting part from its original position as it enters the pelvic inlet (usually OT) to the AP position as it passes through the pelvis. As with flexion, internal rotation is a passive movement that results from the shape of the pelvis and the pelvic floor musculature. The pelvic floor musculature, including the coccygeus and ileococcygeus muscles, forms a V-shaped "hammock" that diverges anteriorly. As the head descends, the occiput of the fetus rotates toward the symphysis pubis—or, less commonly, toward the hollow of the sacrum—thereby allowing the widest portion of the fetus to negotiate the pelvis at its widest dimension. Owing to the angle of inclination between the maternal lumbar spine and pelvic inlet, the fetal head engages in an asynclitic fashion (i.e., with one parietal eminence lower than the other). With uterine contractions, the leading parietal eminence descends and is first to engage the pelvic floor. As the uterus relaxes, the pelvic floor musculature causes the fetal head to rotate until it is no longer asynclitic.

Extension

***Extension* occurs once the fetus has descended to the level of the introitus.** This descent brings the base of the occiput into contact with the inferior margin at the symphysis pubis. At this point, the birth canal curves upward. The fetal head is delivered by extension and rotates around the symphysis pubis. The forces responsible for this motion are the downward force exerted on the fetus by the uterine contractions along with the upward forces exerted by the muscles of the pelvic floor.

External Rotation

External rotation, also known as *restitution,* refers to the return of the fetal head to the correct anatomic position in relation to the fetal torso. This can occur to either side, depending on the orientation of the fetus. This is again a passive movement that results from a release of the forces exerted on the fetal head by the maternal bony pelvis and its musculature and is mediated by the basal tone of the fetal musculature.

Expulsion

***Expulsion* refers to delivery of the rest of the fetus.** After delivery of the head and external rotation, further descent brings the anterior shoulder to the level of the symphysis pubis. **The anterior shoulder is delivered in much the same manner as the head, with rotation of the shoulder under the symphysis pubis.** After the shoulder, the rest of the body is usually delivered without difficulty.

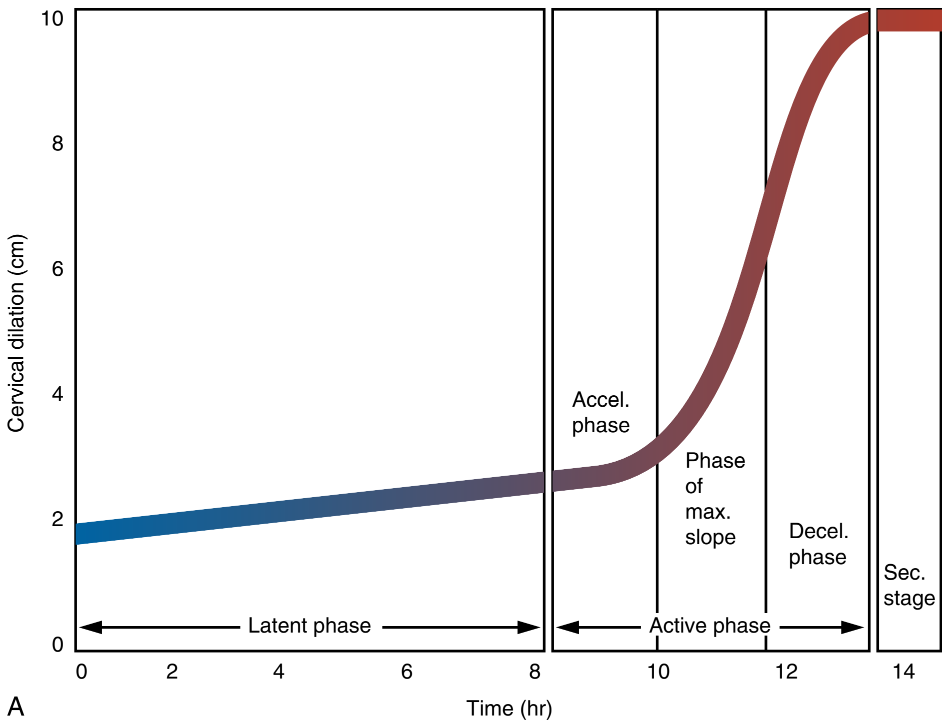
NORMAL PROGRESS OF LABOR

Labor progress is measured with multiple variables. With the onset of regular contractions, the fetus descends in the pelvis as the cervix both effaces and dilates. With each vaginal examination to judge labor progress, the clinician must assess not only cervical effacement and dilation but also fetal station and position. This assessment depends on skilled digital palpation of the maternal cervix and the presenting part. As the cervix dilates in labor, it thins and shortens, or becomes more effaced. Cervical *effacement* refers to the degree of cervical thinning and can be reported in cervical length (CL) or as a percentage. Most clinicians use percentages to follow cervical effacement during labor. Zero percent effacement at term refers to at least a 2 cm long or a very thick cervix, while 100% effacement refers to no length remaining, or a very thin cervix. Generally, 80% or greater effacement is observed in women who are in active labor. *Dilation* ranges from no dilation (closed) to 10 cm (completely dilated or the inability to palpate cervix around the presenting part). For most women, a cervical dilation that accommodates a single index finger is equal to 1 cm, and two index fingers' dilation is equal to 3 cm. The assessment of station, discussed earlier, is important for documentation of progress, but it is also critical when determining whether an OVD is feasible. Fetal head position should be regularly determined once the woman is in active labor; ideally this should occur before significant caput has developed, which obscures the sutures. Like station, knowledge of the fetal position is critical before performing an OVD (see Chapter 15).

Labor occurs in three stages: the *first stage* is from labor onset until full dilation of the cervix; the *second stage* is from full cervical dilation until delivery of the baby; and the *third stage* begins with delivery of the baby and ends with delivery of the placenta. The first stage of labor is further divided into two phases: *latent phase* and *active phase.* The *latent phase* begins with the onset of labor and is characterized by a slow rate of cervical change. Labor onset is a retrospective diagnosis that is defined by the initiation of regular painful contractions of sufficient duration and intensity to result in cervical dilation or effacement. The identification of labor onset often depends on patient memory and the timing of contractions in relation to the cervical examination. When the rate of cervical dilation is accelerated, latent labor ends and active labor begins. **The *active phase* of labor is defined as the period in which the greatest rate of dilation occurs.** Identification of the point at which labor transitions from the latent to the active phase will depend upon the frequency of cervical examinations and retrospective examination of labor progress.

First Stage of Labor

Historically based upon Friedman's[46] seminal data on cervical dilation and labor progress from the 1950s and 1960s, active labor required 80% or more effacement and 4 cm or greater dilation of the cervix. He analyzed labor progress in 500 nulliparous and multiparous women and reported normative data that have been used for more than half a century to define our expectations of normal and abnormal labor.[46,47] Friedman revolutionized our understanding of labor because he was able to plot static observations of cervical dilation against time and successfully translate the dynamic process of labor into a sigmoid-shaped curve similar to that in Fig. 13.13A. Friedman's data popularized the use of the labor graph, which first depicted only cervical dilation and was then later modified to include fetal descent. A cervical dilation of 4 cm marks the transition from the latent to the active phase because it corresponds to the flexion point on the



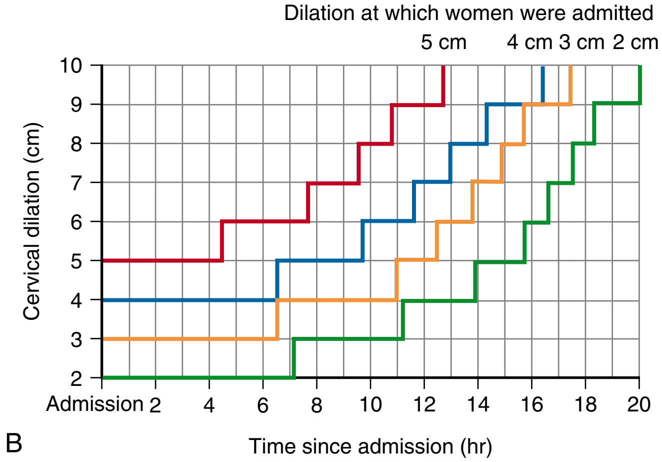


Fig. 13.13 (A) Modern labor graph. Characteristics of the average cervical dilation curve for nulliparous labor. (B) Zhang labor partogram: the 95th percentiles of cumulative duration of labor from admission among singleton term nulliparous women with spontaneous onset of labor. *Accel.*, Acceleration; *Decel.*, deceleration; *Max.*, maximum; *Sec.*, seconds. (A, Modified from Friedman EA. *Labor: Clinical Evaluation and Management*. 2nd ed. Norwalk, CT: Appleton-Century-Crofts; 1978. B, From Zhang J, Landy H, Branch DW, et al. Contemporary patterns of spontaneous labor with normal neonatal outcomes. *Obstet Gynecol*. 2010;1116:1281-1287.)

averaged labor curve generated from a review of 500 individual labor curves in the original Friedman dataset.[46] **Rates of 1.5 and 1.2 cm dilation per hour in the active phase for multiparous and nulliparous women, respectively, represent the 5th percentile of normal.**[46,47] **These data led to the general concept that, in active labor, a rate of dilation of at least 1 cm/h should occur.**

More recent analysis of labor progress from several studies challenged our understanding of the cervical dilation at which active labor occurs and suggested that the transition from the latent phase to the active phase of labor is a more gradual process.[48] An analysis of labor curves for 1699 nulliparous and multiparous women who presented in term spontaneous labor and underwent a vaginal delivery determined that, at a cervical dilation of 4 cm, only half of the women were in the active phase; by 5 cm, 75% of the women were in the active phase; and by 6 cm, 89% of the women were in the active phase.[49] Furthermore, a retrospective review of data from the National Collaborative Perinatal Project, a historic cohort of 26,838 term women in spontaneous labor from 1959 through 1966, used a repeated measures analysis to construct labor curves for parturients whose intrapartum management was similar to those studied by Friedman in the 1950s.[50] **This study determined that labor progress in nulliparous women who ultimately had a vaginal delivery was in fact slower than previously reported until 6 cm of cervical dilation,** after which the rate of labor progress increased. Specifically, most of these women were not in active labor until approximately 5 to 6 cm of cervical dilation. It is important to note that the CD rate in this cohort was 5.6%, and only 20% of nulliparas and 12% of multiparas received oxytocin for labor augmentation.[50]

These findings were confirmed in a prospective analysis of contemporary data collected by the Consortium on Safe Labor, which enrolled and followed 62,415 singleton term parturients who presented in spontaneous labor at 19 institutions from 2002 through 2007.[51] This dataset more accurately reflected a modern obstetric cohort and included a greater percentage of women with oxytocin augmentation (45% to 47%) and epidural analgesia (71% to 84%) compared with those studied by Friedman. Zhang and colleagues[51] reported the median and 95th percentiles of time to progress from one centimeter to the next and confirmed that labor may take more than 6 hours to progress from 4 to 5 cm and more than 3 hours to progress from 5 to 6 cm regardless of parity (Table 13.1). **All women appeared to progress at a similar pace before 6 cm, but multiparas had a faster rate of cervical dilation after 6 cm had been reached.**[51] **These data suggest that it would be more appropriate to utilize a threshold of 6 cm cervical dilation to define the active phase of labor and that the rate of cervical dilation for nulliparas at the 95th percentile of normal may be slower than the 1 cm/h previously expected.** These are important findings, suggesting that clinicians using the Friedman dataset to determine the threshold for active labor and normal labor progress may be diagnosing labor arrest prematurely, which could result in unnecessary CDs.[50-52]

The labor partogram previously in use was based upon the graph introduced in 1964 by Schulman and Ledger.[53] The contemporary data from the Consortium on Safe Labor may be a better way to graphically depict labor progress.[51] **The Zhang partogram** (see Fig. 13.13B) **illustrates the 95th percentile of the duration of labor in hours stratified by cervical dilation on admission. Cervical dilation is not recorded as a continuous measure, and as a result, interval change over time is depicted in a stairstep pattern rather than a continuous sigmoid-shaped curve. The Zhang partogram may be more appropriate for identification of those parturients with a duration of labor that exceeds normal.** Its clinical utility, however, has yet to be confirmed and validated. A Cochrane review of six studies evaluating clinical partogram use failed to show any difference in CD or operative vaginal delivery rate, and it has not been recommended for use in standard labor management.[54]

TABLE 13.1 Median Duration of Time Elapsed (in Hours) for Each Centimeter of Change in Cervical Dilation in Spontaneous Labor Stratified by Parity

	MEDIAN ELAPSED TIME IN HOURS (95TH PERCENTILE)[a]		
Cervical Dilation (cm)	Parity 0	Parity 1	Parity ≥2
3–4	1.8 (8.1)	—	—
4–5	1.3 (6.4)	1.4 (7.3)	1.4 (7.0)
5–6	0.8 (3.2)	0.8 (3.4)	0.8 (3.4)
6–7	0.6 (2.2)	0.5 (1.9)	0.5 (1.8)
7–8	0.5 (1.6)	0.4 (1.3)	0.4 (1.2)
8–9	0.5 (1.2)	0.3 (1.0)	0.3 (0.9)
9–10	0.5 (1.8)	0.3 (0.9)	0.3 (0.8)

[a]An interval-censored regression model was used to estimate the distribution of time for progression from 1 cm to the next with the assumption of log normal distribution of the labor data.
Modified from Zhang J, Landy H, Branch D, et al. Consortium on safe labor: contemporary patterns of spontaneous labor with normal neonatal outcomes. *Obstet Gynecol*. 2010;116:1281-1287.

While Zhang's landmark data[51] reported that the median cervical effacement at admission was 90% for nulliparas, 90% for those women with one previous delivery, and 80% for those with two or more previous deliveries, there is no information about the progress of cervical effacement and its potential impact on the rate of cervical dilation. A 2016 secondary analysis of labor data from a prospective trial of fetal pulse oximetry assessed the association between cervical effacement and rate of cervical change among nulliparous women.[55] In this study, labor data for 488 spontaneously laboring women were compared for those at 100% effacement versus those with less than 100% effacement. In this cohort, the time to dilate from 2 to 3 cm and from 3 to 4 cm was not different by effacement. However, the time to progress between successive dilations after 4 cm was significantly shorter by 16 to 32 minutes for each centimeter of dilation ($P = .01$ to $P < .001$) for those who were 100% effaced compared to those who were not.[55] This suggests that cervical effacement does influence the rate of labor progress, and future studies are needed to confirm these findings. It also supports future labor guidelines considering the inclusion of cervical effacement in defining active labor and labor curves or partograms.

Second Stage of Labor

Because most nulliparous women in Friedman's era had a forceps delivery once the duration of the second stage reached 2 hours, his reported second-stage lengths are somewhat artificial. **Overall, a wide range of mean and 95th percentile second-stage labor durations have been reported for vaginal deliveries.**[46,47,51,56-60] More recent labor duration data that evaluated women in spontaneous labor without augmentation or operative delivery from multiple countries reported similar mean durations, suggesting that these normative data are reliable and useful.[61] These are summarized in Table 13.2 for nulliparas and multiparas with and without epidural analgesia.

TABLE 13.2 Summary of Means and 95th Percentiles for Duration of First- and Second-Stage Labor

Parameter	Mean	95th Percentile
Nulliparas		
Latent labor	7.3—8.6 h	17—21 h
First stage	6—13.3 h	16.6—30 h
Second stage	36—57 min	122—197 min
Second stage, epidural	79 min	336 min
Multiparas		
Latent labor	4.1—5.3 h	12—14 h
First stage	5.7—7.5 h	12.5—13.7 h
Second stage	17—19 min	57—81 min
Second stage, epidural	45 min	255 min

Data from Zhang J, Troendle J, Mikolajczyk R, et al. The natural history of the normal first stage of labor. *Obstet Gynecol*. 2010;115(4):705-710; Zhang J, Landy HJ, Ware Branch D, et al. Contemporary patterns of spontaneous labor with normal neonatal outcomes. *Obstet Gynecol*. 2010;116(6):1281-1287; Rouse DJ, Owen J, Savage KG, et al. Active phase labor arrest: revisiting the 2-hour minimum. *Obstet Gynecol*. 2001;98(4):550-554; and Schulman H, Ledger W. Practical applications of the graphic portrayal of labor. *Obstet Gynecol*. 1964;23(3):442-445.

Factors Affecting Normal Labor Progress

Factors that affect the duration of the first stage of labor include parity, maternal body mass index (BMI), maternal age, fetal position and size, and epidural analgesia.[62-65] Historical data conflict in regard to the effect of epidural use on the duration of the first stage of labor.[56,57,59] Retrospective cohort studies have suggested that epidural use may significantly increase the duration of the first stage of labor compared to the use of opioids only for pain control. In addition, a Cochrane meta-analysis of nine RCTs did identify a statistically significant increase in the mean length of the first stage of labor in women randomized to epidural analgesia compared with those who received opioids only (average mean difference [MD], 32.28 minutes; 95% CI, 18.34 to 46.22).[35] Additional studies are needed to confirm the effect of epidural use on the 95th percentile duration for the first stage of labor.

A retrospective review of the Consortium on Safe Labor database determined median and 95th percentile labor durations and reconstructed labor curves by birthweights.[66] **In nulliparous women, there was a significant increase in the times for each centimeter from 3 to 10 cm for every 500 g increase in birthweight even in normal-weight fetuses** ($P < .01$).[66] This same trend was true in multiparous women, but only statistically significant for each centimeter between 3 and 8 cm.[66] In the reconstructed labor curves, the inflection point representing active labor was consistently 6 cm for all birthweights; interestingly, the time to reach this inflection point increased by birthweight, **suggesting a longer time to reach active labor as birthweight increases.**[66]

Multiple factors influence the duration of the second stage; these include parity, induced labor, longer active phase, ethnicity, older maternal age, maternal BMI and excess maternal weight gain, fetal head position, greater birthweight, and epidural analgesia.[61,64,67-70] A retrospective review of second-stage durations for 4605 women reported a 60-minute increase in the median duration of second-stage labor for nulliparas with an epidural compared with those without.[71] A Cochrane meta-analysis of 16 RCTs confirmed that **epidural use significantly increased the mean duration of the second stage compared with no epidural** (average MD, 15.38 minutes; 95% CI, 8.97 to 21.79) and increased the rate of operative delivery (30 studies: OR, 1.44; 95% CI, 1.29 to 1.60).[35] According to the ACOG/Society for Maternal-Fetal Medicine (SMFM) obstetric care consensus,[72] it is important to consider not just the mean or median duration of the second stage with epidural analgesia but also the 95th percentile duration. In a large retrospective cohort study of 33,239 women with term spontaneous vaginal deliveries, **an epidural was associated with an increase in the 95th percentile duration of the second stage in nulliparous women by 94 minutes ($P < .001$) and in multiparous women by 102 minutes** ($P < .001$).[58] Determination of the upper limits of time for normal second-stage labor that incorporates epidural use and other more contemporary labor interventions is helpful in identifying normative values for labor duration that are associated with the lowest risk of maternal and neonatal morbidity.[56,73,74]

A direct correlation has been found between second-stage duration, adverse maternal outcomes (hemorrhage, infection, perineal lacerations), and the likelihood of a successful vaginal delivery.[72] A secondary analysis of a multicenter study on fetal pulse oximetry compared second-stage duration with maternal and perinatal outcomes in 4126 nulliparous women.[75] Chorioamnionitis (overall rate 3.9%), third- and fourth-degree lacerations (overall rate 8.7%), and uterine atony (overall rate 3.9%) were significantly increased with longer second-stage duration (combined OR, 1.31 to 1.60; 95% CI, 1.14 to 1.86).[75] Adverse maternal outcomes with longer labor duration in multiparous women have also been reported.[76,77] In regard to the correlation between adverse neonatal outcomes and longer second-stage lengths, the reports are mixed. Some studies reported a lack of association between neonatal outcomes and duration of the second stage of labor in nulliparous women.[75,78,79] Additionally, in the only RCT on prolonging the second stage of labor in nulliparous women with epidural anesthesia, increasing the "time limit" for the length of second stage by 1 additional hour (from 3 to 4 hours) resulted in a decreased risk of CD by 55% without an increase in adverse neonatal outcomes.[80] However, in a study of 43,810 nulliparas in the second stage of labor, second-stage lengths of greater than 3 hours were associated with increased maternal and neonatal morbidity.[60] In nulliparous women with an epidural who had a non-OVD, a prolonged second stage greater than 3 hours was associated with an increased risk of a shoulder dystocia (adjusted OR, 1.62; 95% CI, 1.17 to 1.65), 5-minute Apgar score less than 4 (adjusted OR, 2.58; 95% CI, 1.07 to 6.17), neonatal intensive care unit admission (adjusted OR, 1.25; 95% CI, 1.02 to 1.53), and neonatal sepsis (adjusted OR, 2.01; 95% CI, 1.39 to 2.91).[60] It should be noted that, although these results are significant, the absolute incidence of each event is extremely low. These findings suggest that the benefits of vaginal delivery with second-stage prolongation must be weighed against possible small but significant increases in neonatal risk.[72] Although less frequent, adverse neonatal outcomes for multiparous women with a prolonged second-stage duration have also been reported.[76,77]

Based upon the available contemporary evidence of normal labor progress and attempts to reduce the primary CD rate, a workshop convened by the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), ACOG, and SMFM recommended **new definitions for abnormal labor progress or labor arrest in the first and second stages.**[81] ***First-stage arrest*** **occurs after 6 cm of cervical dilation with membrane rupture and no cervical change after 4 hours of adequate contractions or 6 hours of inadequate contractions.**[81] This guideline is the same for nulliparous and multiparous women. Of note, progress in the first stage should not take only dilation into account; cervical effacement and fetal station

TABLE 13.3 Recommendations for 95th Percentile Duration for the Second Stage of Labor

	95th Percentile
Multiparas	
Second stage without an epidural	2 hours
Second stage with an epidural	3 hours
Nulliparas	
Second stage without an epidural	3 hours
Second stage with an epidural	4 hours

Modified from Spong C, Berghella V, Wenstrom K, et al. Preventing the first cesarean delivery: summary of a joint Eunice Kennedy Shriver National Institute of Child Health and Human Development, Society for Maternal-Fetal Medicine, and American College of Obstetricians and Gynecologists Workshop. *Obstet Gynecol*. 2012;120(5):1181-1193.

should also be considered. **For *second-stage arrest,* the guidelines vary by parity and by whether epidural analgesia is used.** Table 13.3 outlines the guidelines for identifying those women with abnormal prolongation or arrest of the second stage.[81] Documentation of patient progress and individualization of care is paramount because **longer durations of pushing may also be appropriate if maternal and fetal conditions permit. These limits should not be used arbitrarily to justify ending the second stage. They can, however, be used to identify a subset of women who require further evaluation.**[82-86] Whereas evidence suggests maternal morbidity is significantly higher in women with a prolonged second stage,[60,75,78,79,84-88] it is important to note that the decision to proceed with an OVD or CD simply to shorten the second stage must be based upon weighing the risks of operative delivery against the risks associated with a prolonged second stage and the likelihood of a successful vaginal delivery.

Interventions to Promote Labor Progress in the First Stage of Labor

Aside from augmentation with oxytocin, various interventions have been suggested to promote normal labor progress; some of these include maternal ambulation and upright maternal positioning, labor support, peanut ball use, water immersion, antispasmodic agents, and IV fluid management.[89] A well-designed randomized trial of over 1000 low-risk women in early labor at 3 to 5 cm cervical dilation compared ambulation with usual care and found no differences in the duration of the first stage, need for oxytocin, use of analgesia, neonatal outcomes, or route of delivery.[90] These results suggest that, given appropriate staffing resources and fetal surveillance protocols, walking in the first stage of labor is an option that may be considered for low-risk women. A Cochrane review of 25 randomized and quasi-randomized controlled trials of considerable heterogeneity found that, in low-risk nulliparous women, upright rather than recumbent positioning during labor was associated with a significantly shorter first stage of labor by 1 hour and 22 minutes for women without epidural anesthesia (average MD, −1.36 hours; 95% CI, −2.22 to −0.51), and a reduction in the risk of CD by 30% (relative risk [RR], 0.71; 95% CI, 0.54 to 0.94).[91] In women with epidural anesthesia the benefit of upright positioning is less clear, as there were similar outcomes of operative delivery and length of labor comparing upright and recumbent positions in a Cochrane review of eight trials and 4464 women.[92] Women should be encouraged to consider upright positioning in labor, and the potential benefits should be discussed to facilitate informed decision making.

Immersion in water is not associated with any change in spontaneous vaginal birth or operative delivery. There was a reduction in pain and the use of regional anesthesia in comparison to no use of water immersion in a Cochrane review of 15 trials and 366 women.[93] Maternal ambulation in women without regional anesthesia results in a shorter overall duration of labor by 3 hours and 57 minutes, and these women have a higher chance of spontaneous vaginal delivery than those who are supine, recumbent, or lateral.[91] Peanut ball use was associated with a nonsignificant decrease in the length of labor of 79 minutes in a meta-analysis of four RCTs and 648 women.[94] Other agents have been trialed, such as antispasmodic medications, including valethamate bromide and hyoscine butylbromide, and resulted in a shorter associated first stage of labor by about 74 minutes compared to either placebo or no medication, but had no effect on the second or third stages of labor.[95]

Another area of interest is the impact of volume and route of fluid intake on labor outcomes.[96] The use of IV fluids is a routine practice in many labor and delivery units, although its benefit compared with oral hydration has not been well elucidated, and the volume and type of IV fluid that may promote optimal progress in labor is unclear. A 2013 Cochrane review of nine randomized trials included studies of considerable heterogeneity.[97] Two trials compared women randomized to receive up to 250 mL/h of lactated Ringers (LR) solution and oral intake versus oral intake alone.[97] No difference was found in the CD rate. However, a reduced duration of labor was reported for women with a vaginal delivery who received LR (MD, −28.86 minutes; 95% CI, −47.41 to −10.30). Four trials compared rates of IV fluids (125 mL/h vs. 250 mL/h) in women whose oral intake was restricted, and a significant reduction in the duration of labor was reported in women who received IV fluid at 250 mL/h.[97] A meta-analysis (seven RCTs, $n =$ 1215) showed that the duration of labor in low-risk nulliparous women may be shortened by about 1 hour with a policy of IV fluids at a rate of 250 mL/h rather than 125 mL/h.[96] A rate of 250 mL/h was significantly associated with a reduction in the incidence of CD compared to 125 mL/h by 30%.[97] In a comparison of the type of IV fluid used in a randomized trial of nulliparas in active labor, IV administration at 125 mL/h of dextrose in normal saline (NS) was associated with a significant reduction in labor length and in second-stage length compared with NS.[98] A meta-analysis of 16 RCTs including 2503 low-risk women showed a decrease of 75 minutes in the length of the first stage of labor in women receiving IV fluids with dextrose compared to IV fluids without dextrose.[99] **The available data suggest that a higher volume of IV fluids per hour (250 mL) with dextrose may be beneficial at reducing first-stage labor length for low-risk women.** Additional studies are needed to determine the optimal volume and type of IV fluid replacement in labor, particularly given the various options for oral nutrition and hydration that are available.

Interventions to Promote Labor Progress in the Second Stage of Labor

A Cochrane review of five RCTs evaluated the effect of any upright position compared with recumbent positioning on route of delivery and labor duration in women with epidural anesthesia in the second stage of labor.[100] No statistically significant difference between groups was observed for second-stage duration, operative delivery, neonatal outcomes, or maternal birth trauma.[100] However, a more recent prospective randomized trial comparing 150 women with epidurals in the second stage who underwent active postural changes to those in the supine position identified significantly more spontaneous vaginal deliveries, fewer OVDs, and fewer CDs in the postural change group.[101] In addition, the total duration of the second stage was significantly shorter in those with postural changes (94.66 minutes vs. 124.30 minutes, $P < .001$).[101] These data suggest that **nonrecumbent positioning in the second stage should be considered, depending upon patient preference** (Video 13.3).

A Cochrane review of 26 trials involved 15,858 low-risk women randomized to continuous labor support (doula) compared with routine care in the first and second stages of labor.[102] Women allocated to a labor support provider were more likely to have vaginal delivery (RR, 1.08; 95% CI, 1.04 to 1.12) compared with those who did not. **A labor support person or doula was associated with a shorter labor and a significant reduction in the use of oxytocin and regional analgesia. Women were less likely to have an OVD** (RR, 0.90; 95% CI, 0.85 to 0.96) **or CD** (RR, 0.78; 95% CI, 0.67 to 0.91) **and endorsed greater personal satisfaction.**[102] In addition, a study evaluating the effects of attending a prenatal childbirth preparation course suggested a shorter duration of first stage of labor in those women who attended the course by a mean difference of 109 minutes ($P = .036$) and also showed higher rates of personal birth satisfaction and breastfeeding.[103] **These data suggest that childbirth preparation and continuous labor support have positive impacts on labor duration and birth outcomes. The data are compelling enough for labor support doulas to receive an "A" rating from the US Preventative Services Task Force, meaning that they improve health outcomes and should be recommended for use during labor.**[104]

Different pushing techniques also influence the duration of the second stage of labor. In the largest RCT on this intervention (one trial, 2414 women), term nulliparous women with epidural analgesia who delayed pushing (waiting at least 1 hour until pushing) had a similar rate of spontaneous vaginal delivery in comparison to early pushing (pushing immediately upon entering second stage).[105] In a meta-analysis (12 RCTs, 5445 women) similar results were noted; however, umbilical cord pH less than 7.1 and suspected sepsis were more common in the delayed pushing group.[106] ACOG is therefore in support of immediate pushing for nulliparous women with epidural anesthesia due to the adverse outcomes associated with delayed pushing and potentially prolonged second stage.[107]

Method of pushing has also been studied in women without epidural anesthesia. In a meta-analysis of 425 patients, pushing with the Valsalva maneuver, compared with open glottis pushing, was associated with a decrease in the duration of labor by 19 minutes, but similar incidences of OVD, perineal laceration, postpartum hemorrhage, and neonatal outcomes.[108] Conventional coached pushing awards the benefit of a shorter second stage by about 13 minutes but no other advantages in the second stage of labor.[109] Coaching assisted by ultrasound (i.e., using visual demonstration by transperineal ultrasound of the progression of fetal head descent) is associated with a shorter active phase of the second stage of labor by about 15 minutes.[110]

Interventions that have not been proven useful in active management of the second stage of labor include maternal stirrup use and fundal pressure (whether manual, belt assisted, or assisted by gentle pressure).[111-115]

Active Management of Labor

Abnormal labor progress and labor arrest due to any reason, referred to as labor dystocia, is the most common indication for CD in nulliparous women and the second most common indication for CD in multiparous women. In the late 1980s, in an effort to reduce the rapidly rising rate of CD, active management of labor was popularized in the United States based on findings in Ireland, where the routine use of active management was associated with very low rates of CD.[116] **Protocols for active management included (1) admission only when labor was established, evidenced by painful contractions and spontaneous rupture of membranes, 100% effacement, or passage of blood-stained mucus; (2) artificial rupture of membranes on diagnosis of labor; (3) aggressive oxytocin augmentation for labor progress of less than 1 cm/h with high-dose oxytocin (6 mIU/min initial dose, increased by 6 mIU/min every 15 minutes to a maximum of 40 mIU/min); and (4) patient education.**[116] Observational data suggested that this management protocol was associated with rates of CD of 5.5% and delivery within 12 hours in 98% of women, and that only 41% of the nulliparas actually required oxytocin augmentation.[116] Multiple nonrandomized studies were subsequently published that attempted to duplicate these results in the United States and Canada.[117-120] Two of these reported a significant reduction in CD when compared with historical controls.[117,119] However, in two of three RCTs, no significant decrease in the rate of CD was observed with active compared with routine management of labor.[121,122] In the third RCT, CD was significantly lower in the actively managed group when confounding variables were controlled.[120] In all randomized trials, labor duration was significantly decreased by a range of 1.7 to 2.7 hours, and neonatal morbidity was not different between groups. In a Cochrane review, a meta-analysis of 11 trials including 7753 women concluded that early **oxytocin augmentation in women with spontaneous labor was associated with a significant decrease in CD** (RR, 0.87; 95% CI, 0.77 to 0.99).[123] A meta-analysis of eight trials (4816 women) determined that **early amniotomy and oxytocin augmentation were associated with a significantly shortened duration of labor** (average MD, −1.28 hours; 95% CI, −1.97 to −0.59).[123] This Cochrane review, in findings similar to prior studies, showed that amniotomy alone did not affect labor length but did increase the rate of CD. In contrast, a subsequent RCT of 144 patients receiving early amniotomy prior to active labor showed a significant reduction in median duration of labor from 364 minutes (201 to 580 minutes) in the routine management group to 235 minutes (117 to 355 minutes) in the early amniotomy group ($P < .001$), and no difference was seen in the rates of CD.[124] This reduction in labor duration may have significant cost and bed-management implications, especially for busy labor and delivery units. **Perhaps the most important factor in the duration of labor for hospitalized patients is delaying admission until active labor has been established.**

SPONTANEOUS VAGINAL DELIVERY

Preparation for delivery should take into account the patient's parity, the progression of labor, fetal presentation, and any labor complications. Among women for whom delivery complications are anticipated (e.g., multiple gestation or the presence of risk factors for shoulder dystocia), transfer to a larger and better-equipped delivery room or the operating room, removal of the foot of the bed, and delivery in lithotomy position may be appropriate. If no complications are anticipated, delivery can be accomplished with the mother in her preferred position. Common positions include the lateral position or the partial sitting position.

The goals of clinical assistance at spontaneous delivery are the reduction of maternal trauma, prevention of fetal injury, and initial support of the newborn. When the fetal head crowns and delivery is imminent, gentle pressure should be used to maintain flexion of the fetal head and to control delivery, potentially protecting against perineal injury. Once the fetal head is delivered, external rotation (restitution) is allowed. During restitution, nuchal umbilical cord loops should be identified and reduced. In cases in which simple reduction is not possible, the body can be delivered through the loop of the nuchal cord and the loop reduced after delivery, or in rare cases, the cord can be doubly clamped and transected. The anterior shoulder should then be delivered by gentle downward traction in concert with maternal expulsive efforts; the posterior shoulder is delivered by upward traction. These movements should be performed with the minimal force possible to avoid perineal injury and traction injuries to the neonatal brachial plexus (Video 13.4).

If possible, the following steps described here are best done with the infant on the mother's abdomen. Initially, **the infant should be wiped dry and kept warm.** Keeping the infant warm is particularly important, and because heat is lost quickly from the head, placing a hat on the infant is appropriate. Routine bulb suctioning of vigorous neonates upon delivery is not associated with any benefits and therefore should not be performed.[125] Secretions or mucus can be cleared if obstructing the airway or if either appears to be copious. **For infants born with meconium-stained amniotic fluid, the ACOG Committee on Obstetric Practice and the American Academy of Pediatrics (AAP) no longer recommend routine intubation and tracheal suctioning.** Rather, the presence of meconium should prompt notification and availability of an appropriately credentialed team. **The vigorous infant may remain on the mother's chest for skin-to-skin contact (SSC) and initial newborn care, while routine resuscitation measures should be performed for the nonvigorous infant.[126]**

The timing of cord clamping historically was dictated by convenience and often performed immediately after delivery. There is approximately 80 to 100 mL of blood that transfers from the placenta in the first minutes following birth in term infants. **New evidence suggests benefits for both preterm and term infants following *delayed umbilical cord clamping*.** A 2019 Cochrane review of 25 RCTs comparing late (at least 30 seconds) to immediate cord clamping in preterm infants at 24 to 36 weeks' gestation showed significant **decreases in neonatal death before discharge and intraventricular hemorrhage (IVH) with late clamping.[127]** Those infants who had their cords clamped later were also noted to have a higher bilirubin level, but no difference was seen in the need for phototherapy between the groups.[127] A 2013 Cochrane review of 15 RCTs that compared late (>1 minute) and immediate cord clamping in term infants showed a significant **increase in infant hematocrit, ferritin, and stored iron at 3 to 6 months** with no significant increase in risk of maternal hemorrhage.[128,129] No increase was seen in neonatal polycythemia or overall rates of jaundice; however, fewer newborns required phototherapy for treatment of jaundice in the immediate clamping group (RR, 0.62; 95% CI, 0.41 to 0.96). In addition, children in the delayed clamping group showed improved scores in social and fine motor skills at 4 years of age, indicating a potential long-term benefit of increased iron stores on developmental outcomes.[130] **In 2020, ACOG issued an updated Committee Opinion affirming the practice of delayed cord clamping for vigorous preterm and term infants for at least 30 to 60 seconds after birth.[131]** There is evidence that a collection of cord blood can occur immediately after delivery while the cord is pulsating for those babies who require cord gas analysis but are undergoing delayed cord clamping.[132]

Once the umbilical cord is clamped and initial newborn assessment is complete, the infant should remain with the mother if at all possible. Early ***skin-to-skin contact*** refers to the placement of the naked infant in a prone position onto the mother's bare chest and abdomen near the breast with the infant's side and back covered by blankets or towels.[133] **Early SSC is recommended for the healthy term newborn immediately after vaginal delivery and as soon as possible after CD** by ACOG, AAP, and the Baby-Friendly Health Initiative developed by the World Health Organization.[134,135] Meta-analysis of data from RCTs (14 trials, 887 participants) suggested that immediate or early SSC increased the likelihood of breastfeeding initiation at 1 to 4 months (RR, 1.24; 95% CI, 1.07 to 1.43) and resulted in higher blood glucose at 75 to 90 minutes of life compared with standard care (MD, 10.49 mg/dL; 95% CI, 8.39 to 12.59).[136] Positive effects of SSC on maternal-infant bonding, breastfeeding duration, cardiorespiratory stability, and body temperature have also been demonstrated.[136-138]

The term ***kangaroo care*** refers to a model of postdelivery care that includes continuous SSC and exclusive breastfeeding for low-birthweight infants; it was originally promoted as an alternative to incubators in resource-poor settings. A meta-analysis of 21 RCTs identified significant reduction in neonatal mortality, nosocomial infection and sepsis, and hypothermia as well as improvements in measures of infant growth, breastfeeding, and mother-infant attachment with kangaroo care compared with conventional methods for low-birthweight infants.[139] Additional prospective RCTs are needed to further characterize the benefits and limitations of SSC and kangaroo care in term and normal-weight infants, in resource-rich settings, and after CD.

THIRD STAGE OF LABOR

Placental separation is heralded by lengthening of the umbilical cord and a gush of blood from the vagina, signifying separation of the placenta from the uterine wall. The third stage of labor is generally short. In a case series of nearly 13,000 singleton vaginal deliveries at greater than 20 weeks' gestation, **the median third-stage duration was 6 minutes and exceeded 30 minutes in only 3% of women.**[140] **Third stages lasting greater than 30 minutes were associated with significant maternal morbidity that included an increased risk of blood loss greater than 500 mL, a decrease in postpartum hematocrit by greater than or equal to 10%, need for dilation and curettage, and a sixfold increased risk of postpartum hemorrhage.**[140,141] **These data suggest that if spontaneous separation does not occur, manual removal or extraction of the placenta should be considered after 30 minutes to reduce the risk of maternal hemorrhage**. This should be performed sooner in the presence of active brisk bleeding. The factors associated with a prolonged third stage of labor include preterm delivery, preeclampsia, labor augmentation, nulliparity, advanced maternal age, and second-stage labor duration greater than 2 hours.[142,143]

The third stage of labor can be managed either passively or actively. Passive management is characterized by clamping of the cord once spontaneous pulsations have ceased and delivery of the placenta by gravity or spontaneously, without manipulation of the uterus or traction on the cord. With passive management, uterotonic medications are not given until after delivery of the placenta. **With active management of the third stage, uterotonic medication is administered shortly after delivery of the baby but prior to delivery of the placenta. Controlled umbilical cord traction and countertraction are used to support the uterus until the placenta separates and is delivered, followed by uterine massage after delivery of the placenta**. Two techniques of controlled traction are commonly used to facilitate separation and delivery of the placenta. In the Brandt-Andrews maneuver, a hand pressed against the abdomen secures the uterine fundus to prevent uterine inversion, while the other hand exerts sustained downward traction on the umbilical cord. With the Credé maneuver, the cord is fixed with the lower hand and, while the uterine fundus is secured and sustained, upward traction is applied by a hand pressed against the abdomen. Care should be taken to avoid evulsion of the cord. Meta-analysis of three RCTs has not shown any maternal benefits for uterine massage when given in isolation without the benefit of oxytocin.[144]

Active management strategies to minimize the risk of postpartum hemorrhage have historically included administration of oxytocin, early cord clamping, controlled traction on the umbilical cord, and fundal massage to facilitate early placental separation.[130,145] A Cochrane meta-analysis of eight randomized to quasi-randomized controlled studies found that for a heterogeneous population of women at mixed risk of bleeding, active management of the third stage of labor was associated with a significant reduction in blood loss over 1000 mL (RR, 0.34; 95% CI, 0.14 to 0.87) and a lower risk of anemia (hemoglobin [Hgb] <9 g/dL; average RR, 0.50; 95% CI, 0.30 to 0.83).[145] However, a significant reduction in neonatal birthweight was identified, which may be due in part to early cord clamping and reduced time for placental transfusion in some patients. In women at low risk for postpartum hemorrhage, active management of the third stage of labor was associated with a significant reduction in total blood loss greater than 1000 mL, reduced need for transfusion, and reduced uterotonic use within the first 24 hours when compared to expectant management.[145] No significant difference in the rate of neonatal jaundice requiring phototherapy (average RR, 1.31, 95% CI, 0.78 to 2.18) was identified. Further, low-risk women who were actively managed were more likely to have postdelivery pain, receive analgesics, and be readmitted for postpartum bleeding compared with women who were expectantly managed. Further study is necessary to better elucidate the etiology behind these findings, particularly in women who are at baseline low risk for bleeding.

A Cochrane review and several additional studies support the role of oxytocin as a key component in the active management strategy to reduce blood loss with delivery.[146-149] ACOG, the World Health Organization, the American Academy of Family Physicians, and the Association of Women's Health, Obstetric and Neonatal Nurses (AWHONN) recommend prophylactic use of a uterotonic (preferably oxytocin) after all births for the prevention of postpartum hemorrhage.[150-153] The National Partnership for Maternal Safety encourages informed decision making regarding the use of prophylactic oxytocin to prevent postpartum hemorrhage in women with spontaneous labor who are without an epidural and are at low risk for hemorrhage.[154] Most recently, the largest Cochrane meta-analysis on postpartum hemorrhage demonstrated that either carbetocin alone, or oxytocin and misoprostol, or oxytocin and methylergonovine (Methergine) are more effective than oxytocin alone for the prevention of postpartum hemorrhage.[155] Carbetocin is associated with the least side effects; unfortunately it is not available in all countries (e.g., United States). Methergine should not be used in women with hypertensive disorders. Alternative uterotonics can be considered, especially in light of recent oxytocin shortages in the United States.

Given the available evidence, at least oxytocin, with consideration of the addition of misoprostol or Methergine as appropriate per patient, or carbetocin where available, should be given to all women upon delivery of the baby, either at the anterior shoulder or full body delivery, but certainly before placental delivery. If the baby is full term and vigorous, delayed cord clamping should then occur while the team performs assessments and keeps the baby warm. Gentle cord traction can be employed to facilitate placental delivery as it has been associated with benefits.

After delivery of the baby and placenta, the placenta, umbilical cord, and fetal membranes should be examined. Placental weight (excluding membranes and cord) varies with fetal weight, with a placental:fetal ratio of approximately 1:6. Abnormally large placentas are associated with such conditions as hydrops fetalis and congenital syphilis. Inspection and palpation of the placenta should include the fetal and maternal surfaces and may reveal areas of fibrosis, infarction, or calcification. Although each of these conditions may be seen in the normal term placenta, extensive lesions should prompt histologic examination. Adherent clots on the maternal surface may indicate recent placental abruption; however, their absence does not exclude the diagnosis. A missing placental cotyledon or a membrane defect suggestive of a missing succenturiate lobe suggests retained placenta and should prompt further clinical evaluation. **Routine manual exploration of the uterus after delivery is unnecessary unless retained products of conception or a postpartum hemorrhage is suspected.**

The site of insertion of the umbilical cord into the placenta should be noted. Abnormal insertions include *marginal insertion,* in which the cord inserts into the edge of the placenta, and *velamentous insertion,* in which the vessels of the umbilical cord course through the membranes before attachment to the placental disc. The cord should be inspected for length (normally about 50 to 60 cm), the correct number of umbilical vessels (normally two arteries and one vein), true knots, hematomas, and strictures. A single umbilical artery discovered on pathologic examination is associated with an increased risk of fetal growth restriction and a higher risk of one or more major congenital anomalies (OR, 6.77; 95% CI, 5.7 to 8.06).[156-159] Therefore, this finding should be relayed to the neonatologist or pediatrician, and any abnormalities of the placenta or cord should be noted in the mother's chart.

EPISIOTOMY AND PERINEAL INJURY REPAIR

Following delivery of the placenta, the vagina and perineum should be carefully examined for evidence of injury. If a laceration is seen, its length and position should be noted and repair should be initiated. Adequate analgesia, either regional or local, is essential for repair. Special attention should be paid to repair of the perineal body, the external anal sphincter, and the rectal mucosa. Failure to recognize and repair rectal injury can lead to serious long-term morbidity, most notably fecal incontinence or fistula formation. The cervix should be inspected for lacerations if an operative delivery was performed or when postpartum bleeding is significant.

Perineal injuries, either spontaneous or with the episiotomy, are the most common complications of spontaneous or operative vaginal deliveries, occurring in one-half to three-quarters of vaginal deliveries. **A *first-degree tear* is defined as a superficial tear confined to the epithelial layer; it may or may not need to be repaired, depending on size, location, and amount of bleeding. A *second-degree tear* extends into the perineal body but not into the external anal sphincter. A tear involving the external anal sphincter is a *third-degree tear*. These are classified by the degree of injury, with a 3a tear involving less than 50% of the sphincter muscle, a 3b tear involving more than 50% of the muscle, and a 3c tear involving the whole muscle and internal anal sphincter. A *fourth-degree tear* extends completely through the sphincter complex and the anal epithelium. All second-, third-, and fourth-degree tears should be repaired** (see Chapter 20). Significant morbidity is associated with third- and fourth-degree tears, including risk of incontinence of flatus or stool, rectovaginal fistula, infection, and pain (see Chapter 15). Primary approximation of perineal lacerations affords the best opportunity for functional repair, especially if rectal sphincter injury is evident, which should be repaired by direct apposition or overlapping the ends and securing them using interrupted sutures (Video 13.5).

***Episiotomy* is an incision into the perineal body made during the second stage of labor to facilitate delivery.** It is by definition at least a second-degree tear. Episiotomy can be classified into two broad categories, midline and mediolateral. With a *midline episiotomy,* a vertical midline incision is made from the posterior fourchette toward the rectum (Fig. 13.14). After adequate analgesia is achieved, straight Mayo scissors are generally used to perform the episiotomy. Care should be taken to displace the perineum from the fetal head. The size of the incision depends on the length of the perineum but is generally approximately half of the length of the perineum and should be extended vertically up the vaginal mucosa for a distance of 2 to 3 cm. Every effort should be made to avoid direct injury to the anal sphincter. Complications of midline episiotomy include increased blood loss, especially if made too early; fetal injury; and localized pain. With a *mediolateral episiotomy,* an incision is made at a 45-degree angle from the inferior portion of the hymenal ring (see Fig. 13.14). The length of the incision is less critical than with midline episiotomy, but longer incisions require lengthier repair. The side to which the episiotomy is performed is usually dictated by the dominant hand of the practitioner (Video 13.6).

Historically, it was believed that episiotomy improved outcome by reducing pressure on the fetal head, protecting the maternal perineum from extensive tearing, and preventing subsequent pelvic relaxation. However, consistent data since the late 1980s have confirmed that **midline episiotomy does not protect the perineum from further tearing, and data do not show that it improves neonatal outcome.**[160,161] **Midline episiotomy was associated with a significant increase in third- and fourth-degree lacerations in nulliparous women with both spontaneous vaginal delivery and OVD.**[161-167] **Episiotomy had the highest odds ratio** (OR, 3.2; 95% CI, 2.73 to 3.80) **for anal sphincter laceration in a large study of nulliparous women when compared with other risk factors, including forceps delivery.**[168] Few papers reported that midline episiotomy was associated with no difference in fourth-degree tears compared with no episiotomy.[169] Randomized trials that compared routine to indicated use of episiotomy report a 23% reduction in perineal lacerations that required repair in the indicated group (11% to 35%).[169] Finally, systematic reviews that compared restrictive to routine use of episiotomy

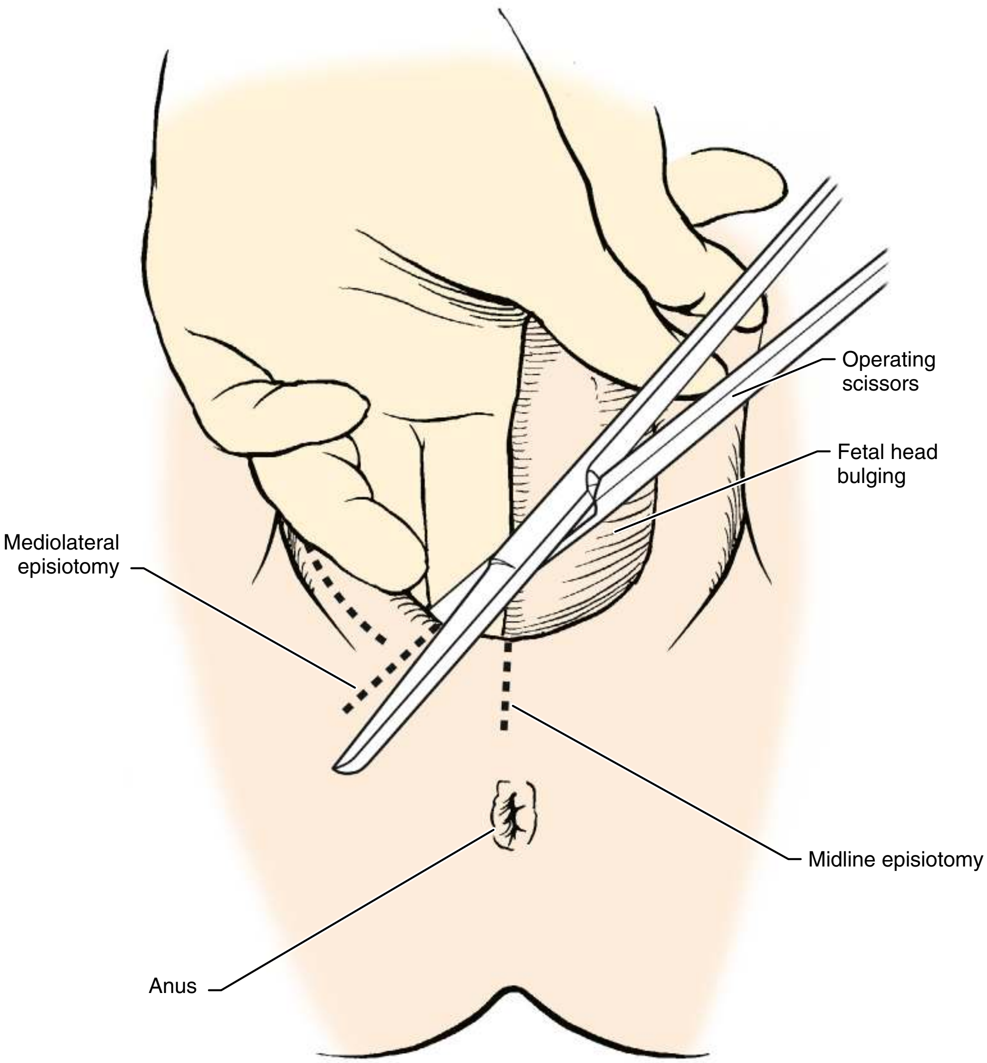


Fig. 13.14 Cutting an Episiotomy.

showed a significant decrease in severe perineal tears, suturing, and healing complications in the restrictive group, as well as no benefit in pelvic floor dysfunction or pelvic organ prolapse with routine use.[170,171] Although the restrictive episiotomy group had a significantly higher incidence of anterior tears, no difference was found in pain measures between the groups. All of these findings were similar whether midline or mediolateral episiotomy was used.

Based on the lack of consistent evidence that episiotomy is of benefit and evidence that it may be more harmful, routine episiotomy has no role in modern obstetrics.[160,161,170,171] Potential indications for episiotomy include a need to expedite delivery in the setting of FHR abnormalities or for relief of shoulder dystocia. Based on these data and the ACOG recommendations,[172] rates of midline episiotomy have decreased, but the procedure is still performed in 10% to 17% of deliveries, which suggests that elective episiotomy continues to be performed.[163-169] In one study, a decreasing episiotomy rate from 87% in 1976 to 10% in 1994 was associated with a parallel decrease in the rates of third- or fourth-degree lacerations (9% to 4%) and an increase in the incidence of an intact perineum (10% to 26%).[163]

The relationship of episiotomy to subsequent pelvic relaxation and incontinence has been evaluated, and no studies suggest that episiotomy reduces the risk of incontinence. Fourth-degree tears are clearly associated with future incontinence,[173] and neither midline nor mediolateral episiotomy is associated with a reduction in incontinence.[174] In fact, a meta-analysis of eight studies **showed an increased risk of anal incontinence with episiotomy** (pooled OR, 1.74; 95% CI, 1.28 to 2.38).[175] If an episiotomy is deemed indicated, the decision of which type to perform rests on their individual risks. It does appear that mediolateral episiotomy is associated with fewer fourth-degree tears compared with midline episiotomy.[176-178] However, other studies do not show benefit of mediolateral over midline episiotomy for future prolapse.[174] Chronic complications, such as unsatisfactory cosmetic results and inclusions within the scar, may be more common with mediolateral episiotomies, and blood loss is greater. Finally, it must be remembered that neither episiotomy type has been shown to reduce severe perineal tears compared with no episiotomy. **Although there is no role for routine episiotomy, episiotomy may rarely be performed in select situations**, and providers should receive training in this skill.[172]

Other methods to reduce the risk of significant perineal lacerations have been studied. These include perineal massage with a water-soluble lubricant in the second stage of labor, which was associated with a 40% increased rate of intact perineum, a 51% decreased rate of severe perineal trauma, and a 44% decreased rate of episiotomy compared with controls (meta-analysis, 3374 women).[179] Perineal hyaluronidase injections during the second stage of labor compared to no intervention showed no clear evidence of a reduction in the incidence of perineal trauma, episiotomy, and third- and fourth-degree perineal lacerations (Cochrane review: four trials, 599 women).[179] In two meta-analyses the application of perineal warm packs during the second stage of labor demonstrated a decrease in third- or fourth-degree lacerations and episiotomy, as well as a higher rate of intact perineum, in comparison with controls (Cochrane review: eight trials, 15,181 women; meta-analysis: seven trials, 2103 women).[180,181] Lastly, the "hands-on" method as originally described by Ritgen,[181a] which includes manual pressure on the fetal head upon crowning at the same time as supporting the perineum with the other hand, was compared to the "hands-poised" technique, in which the fetal head and perineum are not supported by the hands of the accoucheur. Both methods were associated with similar incidences of perineal tears, but the "hands-on" method was associated with a 42% higher incidence of episiotomies.[182] The "hands-on" method was associated with increased risk of third-degree lacerations and did not protect against severe perineal trauma (meta-analysis: five trials, 7587 women).[182]

ULTRASOUND IN LABOR AND DELIVERY

Ultrasound is a useful adjunct to the clinical examination in the peripartum period. Sonographic findings may be used to confirm the clinical impression of fetal lie, presentation, and gestational age when needed. In women with vaginal bleeding, ultrasound can identify placental location and rule out placenta previa prior to a digital examination of the cervix. In women with twins, ultrasound estimation of fetal lie and weight are an integral component of patient counseling regarding mode of delivery for the first and second twin. In women with a breech-presenting fetus at term, ultrasound can be used to confirm presentation, placental location, and amniotic fluid volume prior to patient counseling and performance of an external cephalic version. Postdelivery, ultrasound may be used to assist with removal of the placenta for those women with a prolonged third stage and/or to better facilitate uterine evacuation during a postpartum hemorrhage.

The association between sonographic CL at term and labor outcome has also been studied. An evaluation of weekly CL measurements between 37 and 40 weeks in term nulliparous women found that cervical shortening could only be documented in 50% of the participants prior to onset of spontaneous labor and that 25% of the participants had a CL of more than 30 mm within the last 48 hours prior to delivery.[183,184] CL measurements obtained between 37 and 38 weeks' gestation were found to have a low sensitivity and negative predictive value for spontaneous labor prior to 41 weeks' gestation.[183] In an evaluation of a more ethnically heterogeneous population, however, a significant correlation was identified between a single CL measurement obtained between 37 and 40 weeks' gestation, delivery within 7 days, and delivery prior to 41 weeks' gestation.[185] A term CL of 25 mm had a sensitivity of 77.5% and a negative predictive value of 84.7% for spontaneous labor within 7 days. A CL of 30 mm had a sensitivity of 73.1%, a specificity of 40.7%, and a positive predictive value of 81% for delivery prior to 41 weeks' gestation in the cohort studied.[185] A meta-analysis of transvaginal ultrasound (TVU) CL for prediction of spontaneous labor at term showed that a woman with a singleton gestation at term and a TVU CL of 30 mm has a less than 50% chance of delivering within 7 days, while one with a TVU CL of 10 mm has an over 85% chance of delivery within 7 days (Fig. 13.15).[186] Although these data are interesting and encouraging, additional studies are needed to determine whether term CL evaluation can be used as an adjunct to the clinical examination to better distinguish those women who are most likely to have spontaneous labor between 37 and 40 weeks' gestation.

There is an association between cervical parameters at term and a successful labor induction.[184,185,187] The likelihood of a vaginal delivery following labor induction at term depends in part upon maternal parity and the dilation, effacement, consistency, and position of the cervix.[184] The Bishop score is the most commonly used tool for preinduction cervical assessment. Although it was initially designed to predict the likelihood of vaginal delivery success in multiparous women, the Bishop score has been demonstrated to be applicable to nulliparous women as well.[184,185] The Bishop score is a subjective clinical assessment demonstrated to have significant interobserver and intraobserver variability.[187-189] For women undergoing labor induction, a Bishop score of less than 6 has been reported to be a poor predictor of vaginal delivery and labor induction success.[187] The

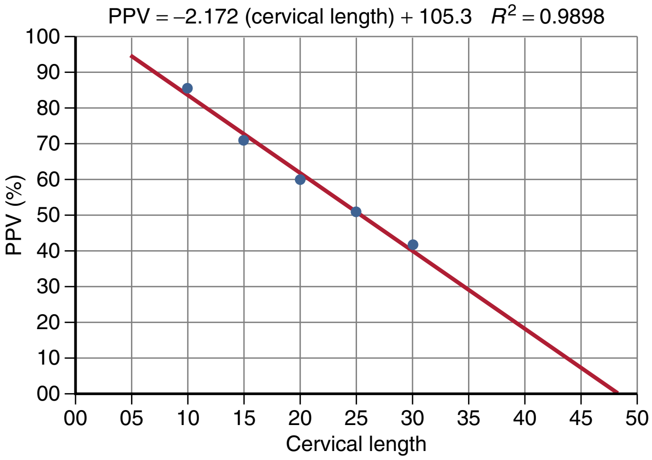


Fig. 13.15 **Association Between Transvaginal Ultrasound Cervical Length at Term and Prediction of Spontaneous Labor Within 7 Days.** Linear regression between cervical length in millimeters (predictor variable; X) and positive predictive value *(PPV)* (criterion variable; Y). (From Saccone G, Simonetti B, Berghella V. Transvaginal ultrasound cervical length for prediction of spontaneous labour at term: a systematic review and meta-analysis. *BJOG.* 2016;123[1]:16-22.)

length of the cervix, an open or closed internal cervical os, and the presence of membranes within the endocervical canal (*funneling,* or *wedging*) are ultrasound parameters that provide a more detailed description of the cervix, particularly when the external os is closed on digital examination. There are conflicting data, however, regarding the association between sonographic cervical parameters and the risk of CD with labor induction.[187,190] A meta-analysis of 20 prospective trials that evaluated the association between preinduction CL and successful labor induction found that a shorter CL was associated with successful induction (likelihood ratio of a positive test, 1.66; 95% CI, 1.20 to 2.31), and a longer CL was associated with a failed induction (likelihood ratio of a negative test, 0.51; 95% CI, 0.39 to 0.67).[191] A meta-analysis of 31 studies, including 12 studies published after 2006, found that in nulliparous women undergoing induction of labor, a CL of more than 30 mm best identified those at high risk for CD, with a sensitivity of 0.70, specificity of 0.74, positive likelihood ratio of 2.7, and negative likelihood ratio of 0.40.[192] A CL of more than 30 mm performed as well as a Bishop score of less than 6 with regard to its sensitivity and specificity for the prediction of a vaginal delivery in nulliparous women.[193] Currently, we do not have sufficient evidence that ultrasound parameters should replace the clinical examination or that such parameters should be used to supplant the indication for induction.

Fetal head position and station during labor are traditionally determined by careful digital pelvic examination. However, evidence suggests that, depending upon the degree of cervical dilation and the presence or absence of caput, an assignment of fetal head position may not be possible in up to 61% of patients in the first stage and in up to 31% of patients in the second stage of labor.[194] An OP fetal head presentation and the presence of caput can also make determination of fetal position and station difficult.[195] Several studies reported that during digital vaginal examinations in which a fetal head position is clinically assigned, the position was accurate only 50% to 60% of the time during the first stage and 30% to 40% of the time during the second stage compared with ultrasound.[195-198] In the second stage of labor, the station of the fetal head within the pelvis is a similarly challenging clinical parameter to assess despite our reliance upon bony pelvic landmarks. In a study using a fetal mannequin and a pelvic trainer, it was found that mid, low, high, and outlet pelvic classifications were incorrectly assigned 30% of the time by resident physicians and 34% of the time by attending physicians.[199]

Ultrasound may be considered as an adjunct to the clinical examination in the second stage, when head position is difficult to assess by palpation because of caput. Ultrasound assessment in labor may be performed using a transabdominal, translabial or transperineal approach, depending on the clinical indication. A two-dimensional ultrasound machine with a convex probe is used, such as a wide-sector probe with low-frequency (<4 MHz) insonation, which is a probe comparable to that used for transabdominal ultrasound for fetal biometry and assessment of anatomy.[200] Sonographic visualization of the orbits will help to determine the general position of the face and occiput within the maternal pelvis. In a study of 514 term nulliparous women who required second-stage OVD, subjects were randomized to predelivery assessment of head position by ultrasound and clinical examination or by clinical examination alone.[201] Head position at delivery was incorrectly predicted by clinical examination 20% of the time compared with a 1.6% error rate when head position was determined by clinical examination and ultrasound ($P < .001$).[201] No statistically significant difference was found between groups in the maternal or neonatal outcomes studied.

Frequently used metrics for intrapartum sonography during labor include the head—perineum distance, the head—symphysis distance, the head direction, and the angle of progression.

Several prospective studies have evaluated the use of ultrasound in predicting the probability of spontaneous vaginal delivery, operative delivery, or CD in patients with a prolonged second stage of labor. Head direction, defined as the angle between the longest axis of the fetal head and the long axis of the pubic symphysis, measured transabdominally in a midsagittal transperineal plane, is a useful tool in predicting delivery.[200] Head direction is categorized as "head down" (angle <0°), "head horizontal" (angle 0° to 30°) and "head up" (angle >30°). Fig. 13.16 presents examples of head horizontal and head up positioning. In 62 women with a prolonged second stage of labor in the OA position, it was found that the head up direction with respect to the pubis symphysis was associated with spontaneous vaginal delivery in the majority of cases (16 of 20, or 80%) compared with the head down (4 of 20, or 20%) or head horizontal (9 of 22, or 41%) directions.[202]

The angle of progression (Fig. 13.17), defined as the angle between the long axis of the pubic bone and a straight line from the posterior edge of the pubis peripheral to the posterior portion of the fetal skull, as measured by transperineal ultrasound, has also been studied in a prolonged second stage of labor. In 41 women with failure to progress in the second stage and the fetus in OA position, when the angle of progression was greater than 120 degrees, the probability of a successful vacuum delivery or spontaneous vaginal delivery was greater than or equal to 90%.[203] In an evaluation of intrapartum ultrasonography for prediction of failed vacuum delivery, an angle of progression less than 105° with pushing, progression distance of less than 25 mm, head down direction, and a midline angle of greater than 45° were predictive of failed OVD.[204]

Another tool that has been studied is the head—symphysis distance, performed transabdominally. This is the distance between the posterior border of the pubic symphysis and the anterior position of the fetal skull, perpendicular to the long axis of the symphysis (Fig. 13.18). This measurement, used during the second stage of labor, was shown to predict the probability of OVD with an accuracy of up to 87.9%, with shorter distances associated with higher success for operative delivery.[205]

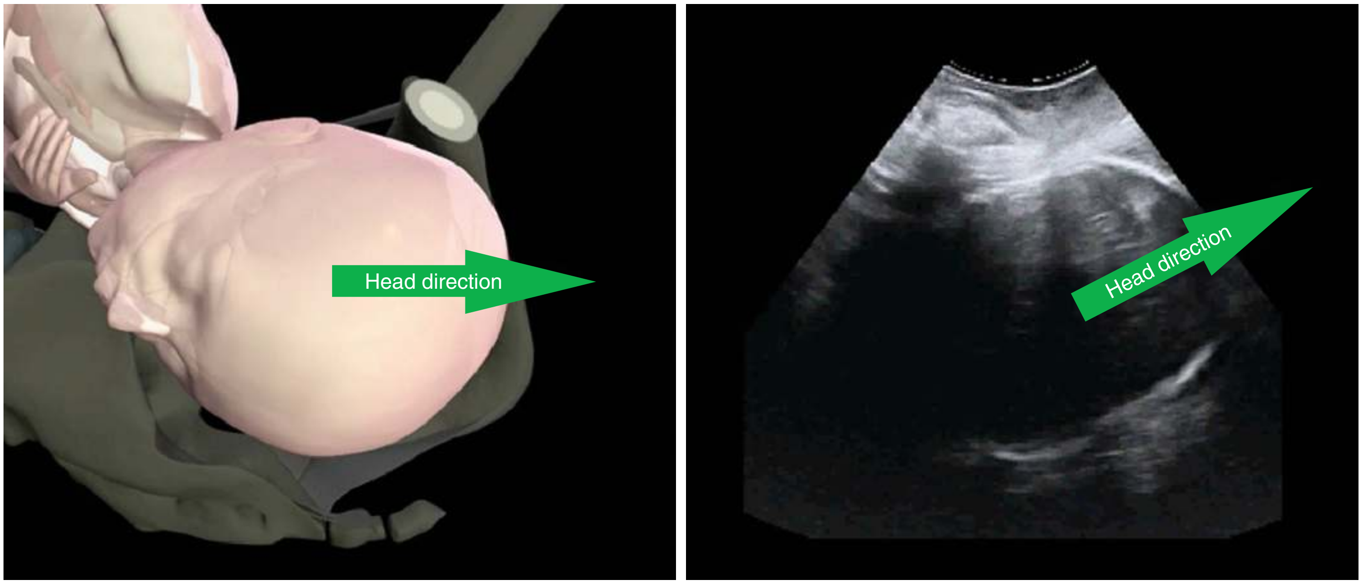


Fig. 13.16 **Fetal Head Direction.** *Left,* Head horizontal direction. *Right,* Head up direction. (From Ghi T, Eggebø T, Lees C, et al. ISUOG Practice Guidelines: intrapartum ultrasound. *Ultrasound Obstet Gynecol.* 2018;52[1]:128-139.)

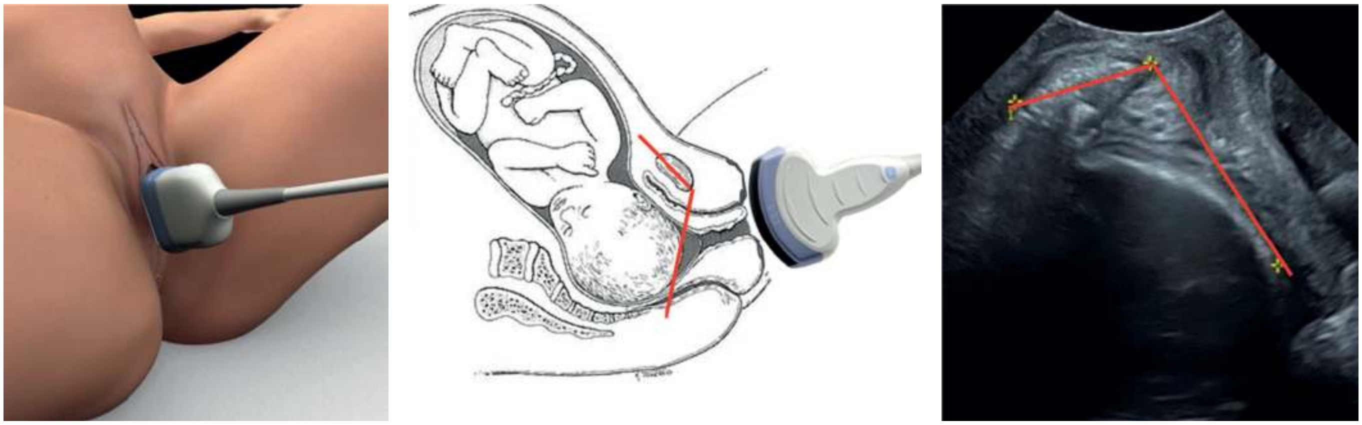


Fig. 13.17 **Measurement of Angle of Progression.** Shown are placement of the transducer *(left)* and how the angle is measured *(center and right).* (Courtesy of A. Youssef, E. A. Torkildsen, T Eggebø and T Ghi.)

Additionally, one study evaluated the superiority of head–perineum distance versus the angle of progression to predict vaginal delivery and time remaining in labor in nulliparous women with prolonged labor.[206] To perform the measurement, the transducer is placed translabially between the labia majora when employing the head–perineum distance measurement. The transducer is next angled to examine the skull shape, with the ultrasound beam perpendicular to the fetal skull. The sonographic marker, called the head–perineum distance, is quantified as the distance from the outer fetal skull to the perineum. This indicates the distance of the part of the vaginal canal not yet passed by the fetus. This study reported that both head–perineum distance (area under the curve 81%) and the angle of progression (area under the curve 72%) were predictive of vaginal delivery.[206] Both methods were also predictive of delivery timing.[206]

Ultrasound can also be used to support the chance of successful manual rotation from OP to OA. Transabdominal ultrasound of the fetal head and spine can straightforwardly differentiate OA versus OP positions. In one RCT of ultrasound examination of nulliparous women with a prolonged second stage with the fetus in OP position, providers who knew the fetal spine position had more successful manual rotations than those who did not know this information (83% vs. 41%). Patients had a significantly higher chance of spontaneous vaginal deliveries (69% vs. 28%) and superior maternal outcome (intact perineum, less blood loss) when fetal spine position was identified prior to delivery.[207]

One additional provocative prospective study of nulliparous women at term in the second stage evaluated transperineal ultrasound volume acquisition at the beginning of active second stage and at 40-minute intervals until delivery.[208] This mechanism is available if three-dimensional ultrasound monitoring is used during labor. Characteristics that were evaluated included the angle of progression, progression distance, head–symphysis distance, and head direction. The study compared both OA and OP fetuses in those women who delivered within 60 minutes of active second stage and those who delivered

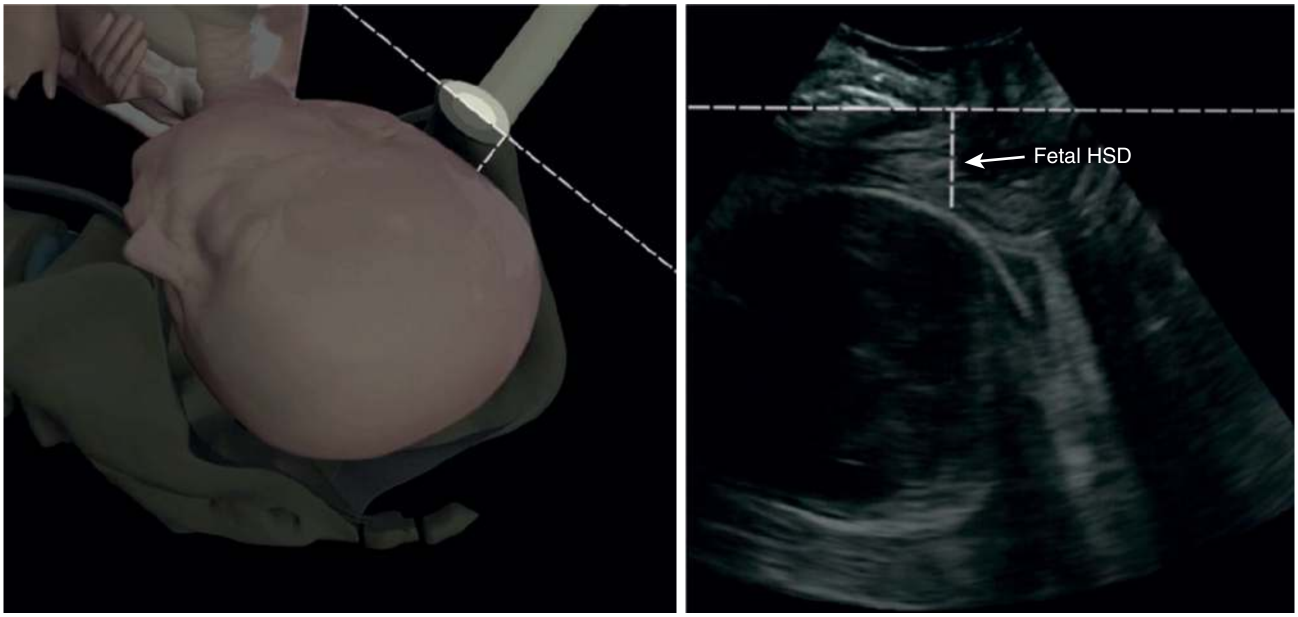


Fig. 13.18 Measurement of head–symphysis distance (HSD), showing placement of transducer and how distance is measured. (From Youssef A, Maroni E, Ragusa A, et al. Fetal head-symphysis distance: a simple and reliable ultrasound index of fetal head station in labor. *Ultrasound Obstet Gynecol.* 2013;41:419-424.)

thereafter. Results showed that in women with normal labor the ultrasound markers of angle of progression (narrower), head–symphysis distance (greater), and progression distance (greater) were statistically significantly different from those markers in women with a prolonged second stage of labor. Additionally, the ultrasound parameters of angle of progression (narrower), head direction (more likely to be head down), and head–symphysis distance (greater) were also different between women with OP fetuses in comparison to women with OA fetuses.[208]

Additional trials are needed to determine whether ultrasound-based evaluations of fetal position and station have sufficient sensitivity to influence clinical decision making regarding second-stage labor and route of delivery.[191,192,201,209]

KEY POINTS

- Labor is a clinical diagnosis that includes regular, painful uterine contractions and progressive cervical dilation and effacement.
- The fetus likely plays a key role in determining the onset of labor, although the precise mechanism by which this occurs is not clear.
- Labor has three stages: the first stage is from labor onset until full dilation of the cervix; the second stage is from full cervical dilation until delivery of the baby; and the third stage begins with delivery of the baby and ends with delivery of the placenta. The first stage of labor is divided into two phases: the first is the *latent phase,* and the second is the *active phase.*
- Active labor is diagnosed as the time when the slope of cervical change increases, which is more difficult to identify in nulliparas and may not occur until at least 6 cm of dilation.
- The ability of the fetus to successfully negotiate the pelvis during labor and delivery is dependent on the complex interaction of three variables: uterine force, the fetus, and the maternal pelvis.
- Labor length is affected by many variables, including parity, epidural use, fetal position, fetal size, and maternal BMI.
- Upright, rather than recumbent, positioning during labor in women with epidural anesthesia was associated with a significantly shorter first stage of labor, less epidural use, and a reduction in the risk of CD by 30%.
- The presence of a labor doula was associated with a significant reduction in the use of analgesia, oxytocin, and operative vaginal delivery or cesarean delivery and an increase in patient satisfaction.
- Intravenous fluids with dextrose may reduce the length of the first stage of labor by 75 minutes.
- Traditional coached pushing and ultrasonographically aided pushing are associated with a decrease in the length of the second stage of labor.
- Delayed cord clamping is associated with both short- and long-term benefits for preterm and term infants.
- Routine episiotomy is associated with a significant increase in the incidence of severe perineal trauma and should be avoided.
- Active management of the third stage of labor—mainly consisting currently of at least giving oxytocin immediately after the baby is delivered and gentle cord traction—was associated with a significant reduction in blood loss greater than 1000 mL and therefore a lower risk of maternal anemia.
- Ultrasound may be a useful adjunct to the clinical examination in the peripartum period and in prediction of spontaneous vaginal delivery.

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